



Antimicrobial resistance: Linking molecular mechanisms to public health impact

Ghazala Muteeb^{a,*}, Raisa Nazir Ahmed Kazi^b, Mohammad Aatif^c, Asim Azhar^d, Mohamed El Oirdi^{e,f}, Mohd Farhan^{f,g,*}

^a Department of Nursing, College of Applied Medical Science, King Faisal University, Al-Ahsa, Saudi Arabia

^b Department of Respiratory Therapy, College of Applied Medical Science, King Faisal, University, Al-Ahsa, Saudi Arabia

^c Department of Public Health, College of Applied Medical Science, King Faisal University, Al-Ahsa, Saudi Arabia

^d NAP Life Sciences; Metropolitan Region, Maharashtra 401208, India

^e Department of Biological Sciences, College of Science, King Faisal University, Al Ahsa, Saudi Arabia

^f Department of Basic Sciences, Preparatory Year, King Faisal University, Al-Ahsa, Saudi Arabia

^g Department of Chemistry, College of Science, King Faisal University, Al Ahsa, Saudi Arabia

ARTICLE INFO

Keywords:

Antimicrobial Resistance
Antimicrobial Stewardship Programs
Multidrug-Resistant Pathogens

ABSTRACT

Background: Antimicrobial resistance (AMR) develops into a worldwide health emergency through genetic and biochemical adaptations which enable microorganisms to resist antimicrobial treatment. β -lactamases (blaNDM, blaKPC) and efflux pumps (MexAB-OprM) working with mobile genetic elements facilitate fast proliferation of multidrug-resistant (MDR) and extreme drug-resistant (XDR) phenotypes thus creating major concerns for healthcare systems and community health as well as the agricultural sector.

Objectives: The review dissimilarly unifies molecular resistance pathways with public health implications through the study of epidemiological data and monitoring approaches and innovative therapeutic solutions. Previous studies separating their attention between molecular genetics and clinical outcomes have been combined into our approach which delivers an all-encompassing analysis of AMR.

Key insights: The report investigates the resistance mechanisms which feature enzymatic degradation and efflux pump overexpression together with target modification and horizontal gene transfer because these factors represent important contributors to present-day AMR developments. This review investigates AMR effects on hospital and community environments where it affects pathogens including MRSA, carbapenem-resistant *Klebsiella pneumoniae*, and drug-resistant *Pseudomonas aeruginosa*. This document explores modern AMR management methods that comprise WHO GLASS molecular surveillance systems and three innovative strategies such as CRISPR-modified genome editing and bacteriophage treatments along with antimicrobial peptides and artificial intelligence diagnostic tools.

Conclusion: The resolution of AMR needs complete scientific and global operational methods alongside state-of-the-art therapeutic approaches. Worldwide management of drug-resistant infection burden requires both enhanced infection prevention procedures with next-generation antimicrobial strategies to reduce cases effectively.

1. Introduction

The recent emergence of antimicrobial resistance (AMR) has been identified as a major challenge to global health and threatens decades of progress in infection management. AMR is defined by the ability of microorganisms such as bacteria, viruses, fungi, and parasites to resist antimicrobial agents, creating significant public health challenges,

especially for low- and middle-income countries. At the center of this dilemma are molecular mechanisms that allow pathogens to evade therapeutic interventions, causing persistent infections, increased mortality rates, and economic impacts. Grasping these molecular foundations is crucial for mitigating the epidemic of AMR. A fundamental molecular mechanism responsible for AMR is the synthesis of enzymes that deactivate antibiotics [1]. For example, β -lactamases facilitate the

* Corresponding authors.

E-mail addresses: graza@kfu.edu.sa (G. Muteeb), mfarhan@kfu.edu.sa (M. Farhan).

<https://doi.org/10.1016/j.slasd.2025.100232>

Received 15 February 2025; Received in revised form 20 March 2025; Accepted 8 April 2025

Available online 9 April 2025

2472-5552/© 2025 The Authors. Published by Elsevier Inc. on behalf of Society for Laboratory Automation and Screening. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

hydrolysis of the β -lactam ring present in penicillins, cephalosporins, and carbapenems, which subsequently leads to their ineffectiveness. Extended-spectrum β -lactamases (ESBLs) and carbapenemases, including blaNDM, blaKPC, and blaOXA-48, have been associated with extensive resistance patterns observed in *Escherichia coli* and *Klebsiella pneumoniae* [2,3]. These enzymes are often encoded by genes on plasmids, allowing for horizontal gene transfer (HGT) and the rapid spread of resistance traits across diverse bacterial populations [4]. Efflux pumps are another important molecular mechanism that contributes to AMR. These membrane-associated protein complexes actively expel antibiotics from bacterial cells, reducing intracellular drug concentrations below therapeutic levels. Some of the well-characterized important efflux systems include MexAB-OprM in *Pseudomonas aeruginosa* as a defense against quinolones, aminoglycosides, and β -lactams [5]. It is established that overexpression of efflux pump genes arises either from mutations of regulatory genes or through the input of mobile genetic elements, implying a relationship between genetic adaptability and the evolutionary process of resistance [6].

Another very important way pathogens resist antimicrobial action is by modifying their target. Methicillin-resistant *Staphylococcus aureus* (MRSA) produces a variant form of penicillin-binding proteins, PBP2a, from the mecA gene that decreases the binding of β -lactams to its target site. In addition, mutations in DNA gyrase and topoisomerase IV genes, gyrA and parC, provide resistance against fluoroquinolones [7]. Such changes at the molecular level reflect how pathogens are in a state of adaptation with respect to selective pressure due to antibiotics. Usually, HGT, which includes transformation, transduction, and conjugation [8], is applied in the transfer of resistance genes. The other mobile genetic elements that serve as vectors include plasmids, transposons, and integrons that aid in transferring the resistance genes among species and genera. Integrons particularly play a key role in constructing gene cassettes that encode resistance determinants, leading to the emergence of MDR strains of bacteria [9]. Incorporation of molecular surveillance techniques such as whole-genome sequencing has provided valuable information on the dynamics of these genetic exchanges and their role in AMR [10].

The implications of these molecular mechanisms are profound for public health. Infections caused by these multidrug-resistant (MDR) pathogens, like CRE and vancomycin-resistant *Enterococcus* (VRE), carry a burden of increased morbidity, mortality, and healthcare cost [11]. These organisms cause problems in treating infections, thus often requiring toxic or suboptimal second-line therapies. Pan-resistant strains that are resistant to all available antibiotics will be an imminent crisis in hospital settings and in immunocompromised populations. To really counter AMR, the gap between molecular research and public health strategy has to be filled [12]. The key will be restriction of the misuse of antibiotics, enhancement of infection control, and discovery of new drugs aimed at targeting mechanisms of resistance. Specifically, some emerging technologies being developed include CRISPR-Cas9-based gene editing and bacteriophage therapy, which would directly target resistance determinants to counter resistant pathogens. In contrast, whole molecular surveillance systems would be necessary to track trends and guide policy interventions [13]. The global burden of AMR is in fact inextricably tied to molecular mechanisms of resistance. Its solution would therefore require a joint effort that incorporates molecular inputs in addition to public health activities. Use of advanced technologies and mitigation of the root cause of resistance can decrease AMR and prolong the efficacy of antimicrobial agents for many generations, as it is the case with other human pathogens [14].

Unlike previous reviews that treat molecular mechanisms of AMR and their public health consequences as separate entities, this review presents a unified analysis linking molecular resistance mechanisms — including enzymatic degradation, efflux pump overexpression, and genetic adaptability — with epidemiological trends and public health burdens across clinical, community, and agricultural settings. Furthermore, it uniquely evaluates emerging surveillance platforms such as

WHO GLASS and novel therapeutic interventions including CRISPR-based genome editing, bacteriophage therapy, and AI-driven diagnostics. This integrative approach offers a comprehensive perspective aimed at informing global policy, guiding clinical practice, and identifying research priorities in the fight against AMR.

1.1. Interplay between molecular processes and issues in public health

AMR is a highly multifaceted global issue that cuts deeply into the molecular adaptations of microbes, overlapping at critical public health concerns [15]. On the molecular level, bacteria utilize many different mechanisms of resistance ranging from enzymatic degradation of drugs, through modification of drug targets, reduction in permeability to active efflux mechanisms that render the antimicrobial agent ineffective [16]. While dissemination through HGT and clonal expansion complicates the attempt to control resistant infections, the public health impact reflects the increase in morbidity, mortality, and the associated costs to health care. More insights into molecular drivers of AMR within a broader context could better help design a way to address the crisis better [17]. In AMR, a quintessential molecular mechanism is the production of β -lactamases, which hydrolyze β -lactam antibiotics, thus making them inoperative. Genes encoding β -lactamases, such as blaNDM, blaKPC, blaVIM, and blaOXA-48, are often located on mobile genetic elements like plasmids and integrative conjugative elements, which facilitate their rapid dissemination among bacterial species [18]. The blaNDM gene, for instance, encodes New Delhi metallo- β -lactamase, which hydrolyzes carbapenems, leading to failures in treatment against life-threatening infections. The public health significance of blaNDM is its worldwide dissemination, which has been linked to increased hospital outbreaks and poor therapeutic alternatives. Molecular surveillance of β -lactamase genes can inform policies of infection control and guide the development of novel inhibitors to counteract resistance [19].

Another significant link between molecular resistance and public health problems is efflux pumps. The drug concentration in bacterial cells is diluted below therapeutic thresholds through the active extrusion of antibiotics by transmembrane protein complexes [20]. Some of the best-studied multidrug-resistant systems include the AcrAB-TolC efflux pump in *E. coli* and the MexAB-OprM system in *Pseudomonas aeruginosa*. These efflux pumps are often overexpressed through mutations in regulatory genes, such as marR or nalC, that increase the survival of bacteria under antibiotic pressure. From a public health point of view, efflux-mediated resistance limits the effectiveness of antibiotics and promotes the emergence of MDR pathogens with a consequence of increased length of hospital stay and mortality. Molecules targeting the efflux pumps may reinvigorate the action of antibiotics and, therefore, clinical benefits [21]. One of the more important broad public health impacts of such molecular features of AMR is changes in drug targets. MRSA has an alternative PBP2a provided by the presence of the mecA gene, which has low affinity for β -lactams; thus, it avoids being inhibited by these drugs [22]. Thus, the mutations in the quinolone resistance-determining regions of the gyrA and parC genes are decisive in the establishment of fluoroquinolone resistance since such drugs are utilized in the therapy of various infections. Such molecular changes profoundly influence the public health condition as they allow resistant bacterial species to persist in the nosocomial environments and also in other community environments. Such mutations are increasingly being reported, thus underlining the need for molecular diagnostics to quickly identify resistance determinants and guide appropriate treatment regimens [23]. Mobile genetic elements play a significant role in AMR through horizontal spread of resistance genes among bacterial populations. For example, integrons serve as genetic platforms that capture and express gene cassettes encoding resistance determinants. Class 1 integrons, frequently co-located with MDR pathogens, have resistance gene arrays encoding aminoglycosides, β -lactams, and sulphonamides among others [24]. Molecular epidemiology research finds that the presence of integrons is distributed highly in clinical and environmental

isolates; this makes them a major contributor to resistance in ecosystems. The use of public health strategies, therefore, monitors the spread of integrons, particularly within agricultural and wastewater environments, forms a vital mechanism to control the AMR crisis [25].

One of the most spectacular examples of how the molecular mechanisms of AMR play out into public health issues is the plasmid-mediated colistin resistance. The *mcr-1* gene encodes a phosphoethanolamine transferase that alters the bacterial lipid A moiety to decrease its binding affinity for colistin and bring about resistance [26]. Inasmuch as colistin is generally considered the last drug for MDR Gram-negative infections, the widespread dissemination of *mcr-1* across the world has sounded all alarms. Molecular tracking of *mcr-1*-positive isolates has proven its presence not only in clinics but also on farms, thus establishing the interlinks between human, animal, and environmental health. A major strategy involved in surveillance and containment of such genes is crucial in preserving the use of last-line antimicrobials [27]. Beyond bacterial resistance, molecular details of AMR also point towards its implications on public health intervention. Acquisition of resistance genes frequently incurs a fitness cost on bacteria and might, therefore, limit their further growth. However, compensatory mutations may enhance fitness and enable resistant populations to outcompete sensitive populations after the exposure to antibiotics. For example, compensatory mutations within the RNA polymerase genes of *Mycobacterium tuberculosis* reduce the fitness costs that come with rifampin resistance, enabling the drug-resistant TB to be maintained. The understanding of molecular evolution of resistance in such pathogens is important to envision the interventions for limiting their transmission and improving treatment outcomes [28]. AMR molecular mechanisms link to the broader public health challenges of infection control, treatment failures, and global health inequities. From the mobilization of resistance genes through mobile elements to the development of target site modifications, each mechanism ties into the complexity of the AMR crisis. There is value in linking basic molecular science with public health activities in antimicrobial stewardship, rapid diagnostics, and surveillance, as they represent a step that could help curb the ravages of AMR. Addressing the resistance at the molecular level and in the social spheres would allow us to develop durable solutions to help retain the efficiency of the present antibiotics and safeguard global health security [29].

This review underlines the urgent need to integrate molecular research with public health initiatives for developing sustainable strategies against AMR. Advanced technologies and evidence-based approaches are essential to preserve the effectiveness of antimicrobials and to tackle one of the most significant public health threats of the modern era.

2. Molecular mechanisms of antimicrobial resistance

2.1. Genetic basis of resistance: mutations, plasmids, and transposons

The foundation of AMR is rooted in microorganisms' ability to acquire and spread genetic alterations that provide advantages for survival in the presence of antibiotics. These genetic alterations occur mainly through three mechanisms: chromosomal mutations, plasmid-mediated resistance, and the movement of transposons [30]. Each of these mechanisms plays a unique and vital role in determining the resistance characteristics of pathogens, highlighting the necessity of understanding their molecular foundations to effectively address AMR.

2.1.1. Mutations

In chromosomal DNA frequently target genes that code for antibiotic-binding sites, metabolic enzymes, or regulatory pathways that influence antimicrobial susceptibility. For example, mutations in the *gyrA* and *parC* genes, which encode DNA gyrase and topoisomerase IV, respectively, modify the quinolone-binding pocket, leading to decreased susceptibility to fluoroquinolones [31]. In *Mycobacterium tuberculosis*,

mutations in the *rpoB* gene, which encodes the β subunit of RNA polymerase, result in resistance to rifampin, a key drug in tuberculosis treatment [32]. Additionally, point mutations in promoter regions can cause overexpression of efflux pumps, as observed in *E. coli* and *Pseudomonas aeruginosa*. These mutations improve bacterial survival when exposed to antibiotics and significantly contribute to the persistence of resistant strains in clinical environments.

2.1.2. Plasmids

Act as mobile genetic elements that enable the horizontal transfer of resistance genes among bacterial communities. They often carry genes that encode β -lactamases (such as blaNDM and blaKPC), aminoglycoside-modifying enzymes (like aac(6')-Ib), and components of efflux pumps [33]. The plasmid-mediated colistin resistance through the *mcr-1* gene illustrates the widespread distribution of resistance factors. Plasmids often contain antibiotic resistance operons within integrons, which improve their ability to adapt and integrate into various bacterial species [34]. The conjugative properties of many plasmids allow them to transfer across species barriers, increasing the spread of MDR phenotypes in both healthcare and environmental contexts [35].

2.1.3. Transposons

Commonly referred to as "jumping genes," serve as another key factor in AMR by facilitating the movement of resistance genes within and between genetic elements. Transposons such as Tn3 and Tn4401 carry important clinical genes, including blaTEM and blaKPC, which allow for their integration into plasmids or chromosomal DNA. Composite transposons typically flank resistance genes with insertion sequences, promoting their mobility and stable incorporation [36]. For example, Tn1546, which contains the *vanA* gene cluster, provides vancomycin resistance in enterococci and has been linked to its horizontal transfer to *Staphylococcus aureus*. Transposons also significantly contribute to co-selection, where resistance genes for multiple antibiotics are transferred simultaneously, complicating treatment strategies [37]. The molecular mechanisms driving AMR—mutations, plasmids, and transposons—underscore the dynamic and adaptable nature of bacterial genomes. These genetic alterations not only allow bacteria to endure antibiotic exposure but also enable the swift spread of resistance traits across populations and ecosystems. Addressing these mechanisms through molecular surveillance, gene-editing technologies, and innovative therapeutic strategies is essential for controlling the spread of resistance and maintaining the effectiveness of antimicrobial agents [38].

2.2. Mechanisms of resistance: efflux pumps, enzymatic degradation (e.g., β -lactamases), and target modification

2.2.1. Efflux pumps

Efflux pumps are transmembrane protein complexes that actively remove antibiotics from bacterial cells, decreasing the concentration of drugs inside and making them ineffective. These pumps are categorized into various families, including the ATP-binding cassette (ABC) superfamily, the major facilitator superfamily (MFS), the resistance-nodulation-division (RND) family, the small multidrug resistance (SMR) family, and the multidrug and toxic compound extrusion (MATE) family [39]. Among these, RND pumps, like MexAB-OprM in *Pseudomonas aeruginosa* and AcrAB-TolC in *E. coli*, are extensively studied for their significant role in MDR [40].

At the molecular level, RND efflux pumps are made up of three components: an inner membrane transporter (e.g., MexB), a periplasmic adapter protein (e.g., MexA), and an outer membrane channel (e.g., OprM). These components work together to create a continuous channel that extends across the bacterial envelope. The energy required for the efflux process comes from the proton motive force, which facilitates the conformational changes necessary to transport substrates, including antibiotics, out of the cell [41]. When efflux pump genes are

overexpressed, often due to mutations in regulatory genes like *mexR* or *acrR*, bacteria can develop increased resistance to a broad spectrum of antibiotics, including β -lactams, fluoroquinolones, and tetracyclines. The ability of these pumps to recognize a wide variety of substrates complicates treatment strategies and contributes to the ongoing challenge of MDR pathogens in clinical environments [42].

2.2.2. Enzymatic degradation

Enzymatic degradation of antibiotics represents one of the most common and effective resistance mechanisms. A notable example is β -lactamases, which specifically target β -lactam antibiotics such as penicillins, cephalosporins, carbapenems, and monobactams by hydrolyzing the β -lactam ring, a crucial structure for their bactericidal function [43]. These enzymes are divided into four molecular classes (A-D) based on their amino acid sequences. Class A enzymes (like KPC) and class D enzymes (such as OXA-48) use serine in their active sites for hydrolysis, while class B β -lactamases, known as metallo- β -lactamases (including NDM, VIM, and IMP), depend on zinc ions for their catalytic function. At the molecular level, β -lactamases interfere with the covalent bond between the β -lactam ring and penicillin-binding proteins (PBPs), which stops the inhibition of bacterial cell wall synthesis. Structural studies indicate that β -lactamases have an active site pocket specifically designed to bind and break the β -lactam ring [44]. Changes in the active site, such as mutations in the omega loop or other critical residues, can expand the range of substrates these enzymes can act on. For instance, ESBLs have mutations that allow them to hydrolyze third-generation cephalosporins like cefotaxime and ceftazidime. The spread of β -lactamase genes, often found on plasmids or within integrons, contributes to the rapid dissemination of resistance among bacterial populations and across different regions [45]. Fig. 1 illustrates how bacteria resist antibiotics. Antibiotic X and Antibiotic Y use porin proteins in the outer

membrane to enter the bacterial cell. However, some antibiotics, like Antibiotic Y, are actively expelled from the cell by efflux pumps, which prevents their accumulation and effectiveness. Antibiotic A targets PBPs in the bacterial cell wall, blocking peptidoglycan synthesis, a vital process for maintaining bacterial cell wall integrity. Antibiotic Z cannot penetrate the outer membrane due to its impermeability, making it ineffective. These mechanisms demonstrate the barriers and resistance strategies that bacteria use to avoid the effects of antibiotics.

2.2.3. Target modification

Target modification refers to structural changes in bacterial proteins or enzymes that antibiotics target, which diminishes their binding affinity and leads to resistance. This mechanism is especially significant in resistance to β -lactams, fluoroquinolones, and glycopeptides. In the case of β -lactams, changes in PBPs—the enzymes that facilitate the cross-linking of peptidoglycan strands in the bacterial cell wall—are a key feature of resistance [46]. For instance, in *Staphylococcus aureus*, the acquisition of the *mecA* gene produces PBP2a, a low-affinity PBP that can withstand β -lactam inhibition, resulting in methicillin resistance [47].

When it comes to fluoroquinolone resistance, mutations in the quinolone-resistance-determining regions of *gyrA* and *parC* modify the structure of DNA gyrase and topoisomerase IV, which decreases the binding of the antibiotic. These mutations typically involve changes at critical residues, such as Ser83 and Asp87 in *GyrA*, disrupting the interaction with fluoroquinolone while maintaining enzymatic function. Likewise, resistance to vancomycin in *Enterococcus faecium* is facilitated by the *vanA* operon, which replaces D-Ala-D-Ala in peptidoglycan precursors with D-Ala-D-Lac [48]. This alteration significantly reduces vancomycin's binding affinity by about 1000-fold, making the antibiotic ineffective. Efflux pumps, enzymatic degradation, and target

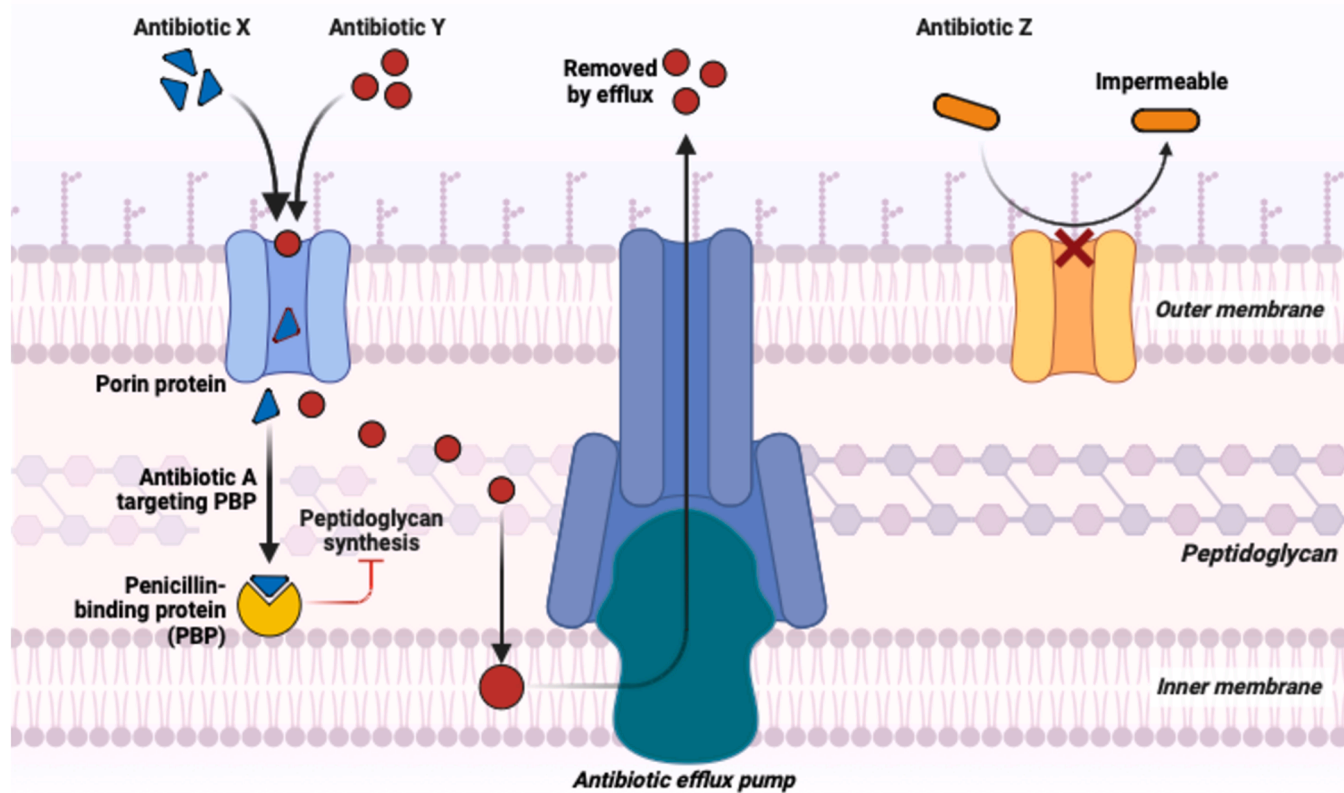


Fig. 1. Mechanisms of bacterial resistance to antibiotics. Antibiotic X and Y pass through porin channels in the outer membrane, while Antibiotic Y is removed by an efflux pump. Antibiotic A disrupts peptidoglycan synthesis by targeting PBPs. Antibiotic Z cannot penetrate the impermeable outer membrane, illustrating resistance.

modification are various molecular strategies that bacteria use to escape the effects of antibiotics. These mechanisms often work together, emphasizing the complexity of AMR and the pressing need for innovative strategies to combat it [49].

2.2.4. Reduced membrane permeability: a critical mechanism of antimicrobial resistance

The multi-dimensional process known as AMR enables bacteria to develop multiple molecular methods that counteract antibiotics. Among fundamental protective mechanisms against antimicrobial agents exists the reduced membrane permeability mechanism in Gram-negative bacteria because these bacteria use their outer membrane barrier effectively to protect themselves against drugs [50]. Gram-negative pathogens use outer membrane porins (OMPs) to control the drug entry process because antibiotics cannot freely cross the peptidoglycan layer which Gram-positive bacteria have. Through porins bacteria allow passive diffusion entry of β -lactams and carbapenems and fluoroquinolones antibiotics. The reduction of antibiotic entry into bacteria cells occurs when porin-encoding genes become mutated or down-regulated thus conferring resistance by blocking drug accumulation inside the cell [51]. Research confirmed the role of mutated OmpK35, OmpK36, and OprD porins in *Klebsiella pneumoniae* and *Pseudomonas aeruginosa* because they develop resistance toward cephalosporins and carbapenems.

The elimination of either ompK35 or ompK36 genes in *Klebsiella pneumoniae* produces cells that have impaired β -lactam penetration resulting in decreased susceptibility to cephalosporins and carbapenems. The absence or structural alterations of porins prohibit antibiotic diffusion thus making clinicians choose between unsafe therapeutic drugs or ineffective options. *Pseudomonas aeruginosa*, with its reputation for resisting multiple antibiotic classes becomes more resistant when it suppresses oprD which serves as a necessary port for carbapenem entry into the bacterium [52]. The resistance in clinical *Pseudomonas aeruginosa* isolates worsens because mutations within the PhoP-PhoQ two-component regulatory system impair OprD suppression. Reduced permeability during bacterial resistance development usually occurs alongside additional resistance mechanisms that compose an effective resistance force toward extreme drug resistance (XDR) phenotypes [53].

The expression of porin at the molecular level adjusts according to environmental factors that include both antibiotic exposure and metabolic nutritional variation. Scientists have proven that changes in ompR and envZ regulatory genes trigger substantial membrane permeability modifications during osmotic and antibiotic stress conditions. Insertion sequence element IS5 has been found above ompK36 which causes both temporary and permanent damage to porin functionality. Reduced membrane permeability caused by porins has extensive clinical effects because as a result β -lactam drugs lose their effectiveness while bacterial resistance to antimicrobial stress increases. The resistance mechanism gets reinforced by carbapenemase production together with blaNDM, blaKPC and blaOXA-48 which results in limited treatment possibilities and complex therapeutic challenges [54].

Molecular surveillance of oprD and ompK variants in hospital settings becomes crucial due to rising porin-deficient resistant strains because it directs effective antibiotic stewardship practices. Researchers investigate new drug methods to restore porins and therapeutic agents with outer membrane permeability-enhancing properties for combating this resistance strategy. Knowledge regarding how membrane permeability interacts with other resistance mechanisms guides the creation of special interventions against multi-drug resistant Gram-negative pathogens. Healthcare officials should integrate genomic and transcriptomic surveillance to monitor reduced permeability changes because this strategy will improve our ability to create future antimicrobial treatments to combat running AMR outbreaks worldwide [55].

2.2.5. Modification of cell membrane charge as a mechanism of antimicrobial resistance

Bacteria have adopted multiple molecular methods to evade antibiotic activity which now represents a crucial worldwide health problem. Bacterial cell membranes use modifications of their electrical charge to protect cells from cationic antimicrobial peptides with polymyxins among them. Gram-negative bacteria cell lysis occurs through polymyxins like colistin (polymyxin E) and polymyxin B after these antibiotics disrupt the lipopolysaccharide (LPS) membrane layer and break membrane integrity. The bacterial pathogens *E. coli* and *Klebsiella pneumoniae* have adjusted their lipid A structure to decrease outer membrane negative charge thus becoming resistant to polymyxin antibiotics. Modification in bacterial composition weakens the electrostatic bonds between polymyxins and bacterial surfaces thereby making these important antibiotics useless [56].

The mobilized colistin resistance (mcr) gene family directs resistance through its phosphoethanolamine transferase enzymes. The phosphoethanolamine transferase enzymes found in LPS activate the lipid A modification process by adding pEtN while decreasing negative charge potential thus blocking colistin from binding. The first identification of mcr-1 gene in plasmids of *E. coli* isolates from livestock during 2015 has led to reports about its appearance in human clinical isolates through HGT [57]. The number of mcr variants (mcr-1 to mcr-10) has been discovered through subsequent investigations while their distinct structural alterations determine their resistance scopes against colistin. Lipid A modification through the action of two-component regulatory system mutations, along with mcr-mediated resistance facilitates increased antibacterial agent resistance.

Healthcare providers face a major challenge due to mcr-positive bacteria because colistin functions as one of the final options for treating carbapenem-resistant *Enterobacteriaceae* (CRE). XDR and pan-drug-resistant (PDR) pathogens have arisen from the combination of mcr genes with carbapenemase genes (blaNDM, blaKPC, blaOXA-48) on conjugative plasmids thus reducing treatment possibilities. Environmental reservoirs that include wastewater and poultry farms along with aquaculture facilities function as key locations for mcr gene transfer because these areas enable the inter-species spread of colistin resistance [58]. Genomic surveillance research demonstrates that mcr-harboring plasmids typically contain insertion sequence elements such as ISAp11 which help plasmids integrate successfully into bacterial genomes to quickly spread the colistin resistance across the world.

A solution to colistin resistance would require implementing molecular surveillance and specific antimicrobial stewardship programs along with creation of new therapeutic approaches. The combination of whole-genome sequencing (WGS) with metagenomics methods gives immediate access to data about mcr-positive strain spread patterns for quick resistance clone detection along with containment methods. The potential therapeutic strategies for reverse colistin resistance via membrane restoration include LPS-binding peptides and anti-resistance adjuvants together with CRISPR-Cas9-based gene editing [59]. The emerging mcr variants need urgent global attention to stop their widespread transmission and protect the effectiveness of antibiotics which serve as last medical options.

2.2.6. Biofilm formation as a molecular strategy in antimicrobial resistance

Biofilms act as an essential molecular system for bacterial pathogens to avoid antimicrobials leading to extended infection periods and increased duration. A biofilm represents bacterial organized communities that exist in a protective self-made extracellular polymeric substance (EPS) matrix which shields them from antimicrobial agents and host immune system interactions. Biofilms possess different bacterial physiological conditions than free-swimming bacteria because bacteria within these communities show different metabolic functions and expression levels of cellular genes and higher rates of horizontal genetic transfer which amplify AMR [60]. Biofilm-associated resistance mechanisms enable *Staphylococcus aureus* together with *Pseudomonas*

aeruginosa and *Enterococcus faecium* pathogens to create persistent infections throughout medical prosthetic devices, catheter-related bloodstream conditions and persistent wound infections in clinical settings.

Biofilm development at the molecular level depends on tight regulation through both quorum sensing systems and internal second messenger cyclic-di-GMP. The lasI-lasR and rhlI-rhlR QS systems in *Pseudomonas aeruginosa* synchronize EPS production of alginate, Pel and Psl which enhances biofilm structural complexity and antibiotic resistance capacities. *Staphylococcus aureus* bacteria use two regulatory systems to control biofilm dispersal through the agr accessory gene regulator and through ica operon enzymes which create poly-N-acetylglucosamine biofilm material [61]. Biofilm-associated bacteria show increased expression of MexAB-OprM efflux pumps together with stress response systems (for example MexAB-OprM in *P. aeruginosa*) which leads to reduced antibiotic susceptibility. Metabolically dormant persister cells forming within biofilms create an extra challenge for antibiotic treatment as they enable bacterial survival during therapy which allows them to reseed infection sites after treatment [62].

Therapy that targets biofilm-associated pathways becomes an effective strategy against AMR due to biofilms' significant involvement in antimicrobial resistance. Biofilm resistance can be mitigated through three evolving strategies that include stopping quorum sensing with small-molecule inhibitors combined with biofilm-dispersing enzymes DNases and dispersin B and anti-biofilm peptides. New drug delivery methods through antibiotic-carrying nanostructures provide better access to biofilm areas which improves infection-fighting capabilities. Knowledge of biofilm regulatory mechanisms at the molecular level demonstrates the need to develop innovative strategies that fight infections caused by biofilm because it represents a critical problem in overcoming antibiotic resistance in medical and environmental environments [63].

2.2.7. Persister cell formation as a molecular strategy in antimicrobial resistance

Bacterial populations develop persister cells as an important non-genetic antimicrobial resistance mechanism to survive antibiotic treatment. The small bacterial subpopulation that transforms into dormant cells functions as persister cells to make them antibiotic-resistant since their resting state decreases their cellular metabolic activity. Persisters enable bacteria to survive antibiotic drugs in chronic infections from pathogens like *Staphylococcus aureus* and *Pseudomonas aeruginosa* along with *E. coli* by undergoing dormant states which ultimately trigger recurring infections [64]. Persister cells develop under antibiotic stress through the control of molecular processes using toxin-antitoxin (TA) systems combined with stringent response pathways as well as ATP depletion mechanisms which establish sustained dormancy.

Multiple persister cell-forming TA systems include hipBA from *E. coli* and mazEF from *S. aureus* at the molecular level. The systems operate through growth arrest control through essential component degradation including mRNA and ribosomes destruction. The alarmone molecule (p) ppGpp participates in bacterial transcriptional regulation through stringent response by reducing gene transcription of metabolism and cell division pathways to boost antibiotic resistance. Lower intracellular ATP levels act as a major determinant for persister formation because they obstruct antibiotic entry while weakening the effectiveness of energy-dependent therapeutic drugs. The existence of persister cells creates a major hurdle for infection treatment because standard antibiotics cannot eliminate dormant cells requiring new therapeutic solutions [65].

The fight against persister-mediated AMR uses metabolism-activating compounds such as sugar analogs and ROS-inducing agents that compel dormant persister cells into a sensitive state by removing dormancy. The strategy of using anti-persister molecules for both TA systems regulators and stringent response elements shows potential for eliminating persistent infections. Research focused on understanding persister cell molecular development is essential for antimicrobial

strategy creation because targeted attacks on these survival populations would improve disease response rates and protect against infections caused by pathogen drug resistance [66] (Fig. 2).

2.3. Horizontal gene transfer and the spread of resistance genes

HGT plays a crucial role in how bacteria obtain and share antibiotic resistance genes among different species and even genera. Unlike vertical gene transfer, which happens during cell replication, HGT allows for the exchange of genetic material between unrelated bacterial cells, enabling them to quickly adapt to the pressures of antimicrobial agents. The main molecular mechanisms of HGT include conjugation, transformation, and transduction, each contributing in its own way to the dissemination of resistance genes [67].

2.3.1. Conjugation

Conjugation involves the transfer of genetic material through direct contact between cells, typically facilitated by plasmids or conjugative transposons. At the molecular level, this process is driven by type IV secretion systems (T4SS) that are encoded by conjugative plasmids, such as the F plasmid found in *E. coli* [68]. The process starts when the donor cell forms a pilus to connect with a recipient cell. A relaxase enzyme then cuts the plasmid's origin of transfer (oriT), which triggers the transfer of a single-stranded DNA through the T4SS. This strand is then replicated in the recipient cell, resulting in a complete plasmid. Plasmids frequently carry resistance genes, including those that code for β -lactamases (like blaNDM and blaCTX-M), aminoglycoside-modifying enzymes, or efflux pumps. Additionally, mobile genetic elements such as insertion sequences (like IS26) and integrons found within plasmids enhance their capacity to capture and move ARGs. Multidrug-resistant

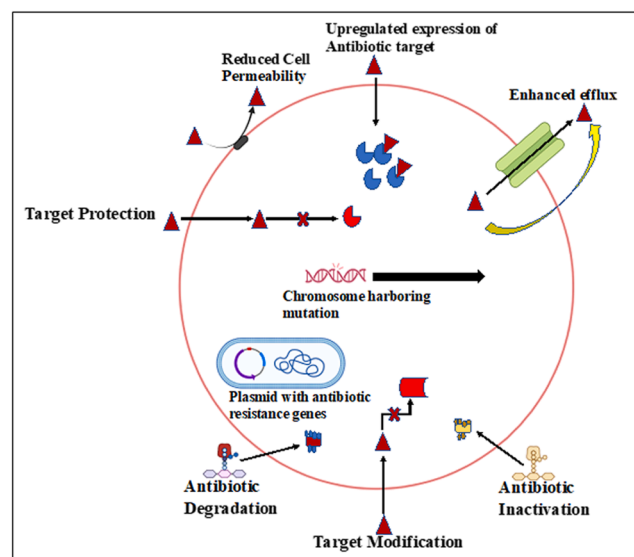


Fig. 2. Mechanisms of Antibiotic Resistance in Bacteria. The figure presents essential processes that enable bacteria to gain resistance against antibiotics. The first resistance mechanism involves (1) Reduced Permeability through which membrane porins undergo mutations or modifications that restrict antibiotic access while maintaining low drug intracellular levels. Bacteria achieve enhanced efflux by operating active pumping mechanisms that remove antibiotics from cells thus lowering their internal amounts. The process of Target Modification arises from chromosomal mutations which modify antibiotic-receptor sites and makes them resistant to antibiotic action. Bacteria create antibiotic resistance through Antibiotic Inactivation which depends on enzymes that both modify and degrade antibiotics until their effectiveness decreases. The enzymes encoded on plasmids that host antibiotic resistance genes accomplish Antibiotic Degradation for helping horizontal gene transfer. Multiple bacterial defense systems exist to guarantee survival during antibiotic therapies making infection treatment more complex.

plasmids, such as the IncF and IncA/C types, illustrate how conjugation speeds up the horizontal transfer of resistance among bacterial populations in both clinical and environmental settings [69].

2.3.2. Transformation

Transformation is the process where bacteria take up and incorporate naked DNA from their surroundings into their genome. Some bacteria, like *Streptococcus pneumoniae* and *Neisseria gonorrhoeae*, are naturally competent and possess specialized molecular machinery to aid in this process. Competence proteins, including ComA and ComK, help in the uptake of double-stranded DNA, which is then processed into single strands by nucleases such as RecA [70]. This single-stranded DNA can integrate into the bacterial chromosome through homologous recombination. Transformation enables bacteria to gain resistance genes from lysed cells or extracellular DNA found in biofilms. For instance, when bacteria take up DNA that encodes variants of penicillin-binding proteins (like PBP2x in *S. pneumoniae*), they develop resistance to β -lactams [71]. Likewise, transformation allows for the acquisition of macrolide resistance genes, such as *ermB*, which code for ribosomal methylases that block antibiotic binding.

2.3.3. Transduction

Transduction refers to the transfer of genetic material through bacteriophages, which are viruses that infect bacteria. In generalized transduction, lytic phages inadvertently package fragments of bacterial DNA, including antibiotic resistance genes, into their capsids. These phages then go on to infect other bacteria, delivering the resistance genes [72]. Specialized transduction happens when temperate phages integrate into the bacterial genome as prophages and later excise, carrying nearby resistance genes into new bacterial hosts. A well-known example of transduction-related resistance is the spread of the *mecA* gene, which encodes PBP2a in MRSA and is linked to the SCCmec cassette that is mobilized by bacteriophages. Additionally, resistance to aminoglycosides and tetracyclines has been associated with the phage-mediated transfer of genes such as *aadA* and *tetM* [73].

2.4. Impact of HGT on resistance gene spread

HGT allows bacteria to quickly adapt to antibiotics, often through the formation of resistance islands or multidrug-resistant integrons, such as class 1 integrons. Integrons are capable of capturing and expressing antibiotic resistance genes through site-specific recombination, which is facilitated by the integrase gene (*intI*). These genetic structures, along with plasmid-mediated conjugation, significantly boost the resilience of bacterial populations facing antimicrobial challenges [74]. The complex interactions between HGT mechanisms and selective pressures in both clinical and environmental contexts highlight the global issue of AMR. To combat HGT, strategies may include targeting the formation of conjugative pili, disrupting competence systems, and employing phage therapy to counteract transduction, emphasizing the necessity for innovative approaches to reduce the spread of resistance genes [75].

3. Epidemiology of antimicrobial resistance

3.1. Prevalence and distribution of AMR in different regions

The global landscape of AMR is influenced by a variety of molecular mechanisms and regional differences in healthcare practices, antibiotic usage, and environmental conditions. The molecular epidemiology of AMR reveals that certain resistance determinants are more common in specific geographical areas, driven by the spread of resistant strains, mobile genetic elements, and various selective pressures [15].

3.1.1. Asia: the hotspot for carbapenemase genes

In Asia, the high rates of AMR are largely due to the widespread presence of carbapenemase-producing *Enterobacteriaceae* (CPE) and

other Gram-negative bacteria. Molecular surveillance has pinpointed blaNDM, blaKPC, and blaOXA-48-like as the main carbapenemase genes. For example, blaNDM is particularly common in India and its neighboring countries, often found on IncF and IncA/C plasmids that promote interspecies transfer [76]. WGS has shown that *Klebsiella pneumoniae* strains carrying blaNDM frequently possess additional resistance genes, such as blaCTX-M, leading to MDR. In Southeast Asia, the proliferation of mobile genetic elements like integrons and transposons (for instance, Tn4401 carrying blaKPC) worsens the AMR situation. Surveillance studies have also detected colistin resistance in *E. coli*, mediated by the *mcr-1* gene, which is often associated with plasmids that move between animal, human, and environmental reservoirs, highlighting the One Health challenge in the region [77].

3.1.2. Europe: dominance of MRSA and VRE

In Europe, MRSA and VRE play major roles in the AMR landscape. Molecular analysis has revealed that the *mecA* gene, which encodes penicillin-binding protein 2a (PBP2a), is crucial for MRSA. The prevalence of MRSA varies across the continent, with higher rates observed in Southern and Eastern Europe compared to the North [78]. Epidemiological studies have connected the spread of MRSA to the SCCmec mobile element, with types IV and V linked to community-acquired MRSA, while types I-III are more common in hospital-acquired infections. The prevalence of VRE is influenced by the *vanA* and *vanB* operons, which modify the D-Ala-D-Ala terminal of peptidoglycan precursors to a D-Ala-D-Lac configuration, thereby reducing vancomycin binding. Genomic studies have traced the dissemination of *vanA* plasmids in healthcare settings, identifying clonal expansions of *Enterococcus faecium* ST80 and ST203 as significant contributors [79].

3.1.3. Africa: emerging resistance genes

In Africa, the lack of extensive molecular surveillance has limited our understanding of AMR. Nevertheless, research indicates an increase in ESBL-producing *Enterobacteriaceae*. The blaCTX-M-15 gene is particularly prevalent, often found alongside mobile genetic elements like IS26 and integrons that facilitate regional spread [80]. Phylogenetic studies have connected these strains to global epidemic lineages, including *E. coli* ST131. There are also high levels of tetracycline resistance, driven by plasmid-borne *tetA* and *tetB* genes, as well as sulfonamide resistance linked to *sul1* and *sul2*, underscoring the impact of mobile resistance factors. The situation is worsened by the common use of substandard antibiotics and inadequate infection control measures in healthcare environments [81].

3.1.4. Americas: multidrug-resistant gram-negative pathogens

In the Americas, the rise of multidrug-resistant Gram-negative bacteria, especially carbapenem-resistant *Pseudomonas aeruginosa* and *Acinetobacter baumannii*, poses significant challenges. Molecular investigations have identified that blaVIM and blaIMP carbapenemase genes are prevalent in Latin America, frequently found alongside genes for aminoglycoside-modifying enzymes (e.g., *aac(6)-Ib*) [82]. The high occurrence of multidrug-resistant *A. baumannii* is associated with global clones (GC1 and GC2) that contain resistance islands with blaOXA-23 and blaOXA-58. In North America, resistance genes like blaKPC have been thoroughly examined, with IncN and IncF plasmids playing a key role in their transmission within healthcare-associated infections. Furthermore, the spread of *mexAB-oprM* efflux pump genes in *P. aeruginosa* illustrates the molecular underpinnings of intrinsic resistance [83].

3.1.5. Global perspective: AMR as a molecular pandemic

The global landscape of AMR is shaped by molecular epidemiology and the movement of genetic elements. Resistance genes such as blaNDM, blaCTX-M, *mecA*, and *mcr-1* highlight the worldwide spread of these resistance factors. Incorporating molecular surveillance techniques, including WGS and metagenomics, into AMR monitoring is

essential for detecting new resistance genes and tracking their distribution across the globe. Tackling AMR necessitates a multidisciplinary strategy that addresses the molecular causes of resistance while also considering regional differences in antibiotic usage and infection control measures [84].

3.2. Key resistant pathogens

3.2.1. *E. coli*

E. coli plays a significant role in AMR, largely due to its capacity to acquire and spread resistance genes. The main resistance mechanism in MDR *E. coli* is the production of ESBLs such as blaCTX-M, blaTEM, and blaSHV [85]. These genes, typically found on plasmids, break down third-generation cephalosporins and aztreonam, making these antibiotics ineffective. Furthermore, the emergence of plasmid-mediated colistin resistance through the mcr-1 gene poses a serious threat, as it modifies the lipid A component of lipopolysaccharides, decreasing colistin's binding ability. Another resistance mechanism in *E. coli* is the overexpression of efflux pumps like AcrAB-TolC, which helps expel various antibiotics, including fluoroquinolones and tetracyclines. High-level fluoroquinolone resistance is also conferred by point mutations in the gyrA and parC genes, which encode subunits of DNA gyrase and topoisomerase IV, reducing the drugs' affinity for their targets. The role of integrons and transposons in disseminating these resistance factors highlights the intricate molecular landscape of AMR in *E. coli* [86].

3.2.2. *Klebsiella pneumoniae*

Klebsiella pneumoniae is a major cause of infections acquired in hospitals and has developed resistance mechanisms that classify it as a high-priority pathogen. The primary means of carbapenem resistance is through carbapenemase enzymes, including blaKPC, blaNDM, and blaOXA-48, which break down carbapenems and other β -lactams. These resistance genes are frequently found on mobile genetic elements such as Tn4401 transposons or IncF and IncX plasmids, aiding their spread worldwide [87]. Hypervirulent strains of *K. pneumoniae* carry both resistance and virulence plasmids, complicating treatment options. For example, the presence of both blaKPC and siderophore genes (like yersiniabactin and aerobactin) in clonal complex 258 (CC258) strains illustrates the overlap of resistance and virulence. Moreover, the increased activity of efflux pumps such as AcrAB-TolC and mutations in outer membrane porins (like OmpK35 and OmpK36) further enhance the MDR characteristics of *K. pneumoniae* [88].

3.2.3. *Staphylococcus aureus*

MRSA poses a significant public health threat due to its resistance to β -lactam antibiotics. This resistance is mainly due to the mecA gene, which encodes penicillin-binding protein 2a (PBP2a), an enzyme that has a lower affinity for β -lactams [89]. The mecA gene resides on the staphylococcal cassette chromosome mec (SCCmec), a mobile genetic element that allows for horizontal gene transfer among *S. aureus* strains. Additionally, resistance to glycopeptides like vancomycin has developed through the acquisition of the vanA operon from Enterococcus species [90]. This operon alters the D-Ala-D-Ala target site of vancomycin to D-Ala-D-Lac, significantly diminishing the drug's effectiveness. MRSA also shows resistance to macrolides, lincosamides, and streptogramins due to the methylation of 23S rRNA by erm genes, which further complicates treatment strategies [91].

3.2.4. *Pseudomonas aeruginosa*

Pseudomonas aeruginosa is an opportunistic pathogen known for its complex resistance mechanisms, making it particularly difficult to treat. It has an impermeable outer membrane and active efflux pumps, such as MexAB-OprM, which can expel a wide variety of antibiotics [92]. Mutations in regulatory genes like nalC and nalD can cause these efflux systems to be overexpressed, further increasing resistance. Additionally,

P. aeruginosa often acquires resistance through the production of β -lactamases, including metallo- β -lactamases (MBLs) such as blaNDM, blaVIM, and blaIMP [93]. These enzymes break down carbapenems and other β -lactams, limiting treatment options. Resistance to aminoglycosides is also a concern, arising from aminoglycoside-modifying enzymes (AMEs) and mutations in ribosomal target sites. The alteration or loss of porin channels, like OprD, adds to carbapenem resistance, emphasizing the complex nature of AMR in *P. aeruginosa* [94].

3.2.5. Global perspective on key resistant pathogens

These resistant pathogens highlight the molecular diversity and adaptability of AMR mechanisms. The combination of plasmid-mediated gene transfer, overactive efflux pumps, mutations at target sites, and enzymatic degradation points to the urgent need for molecular surveillance and innovative treatment strategies. Gaining a deeper understanding of the molecular basis of resistance in these pathogens is essential for creating targeted interventions to address the global AMR crisis [5].

3.3. AMR trends in healthcare, community, and agricultural settings

AMR has become a pressing global public health issue, with its prevalence varying significantly across healthcare, community, and agricultural settings. In healthcare environments, the inappropriate use and overprescription of broad-spectrum antibiotics have led to the rise of multidrug-resistant pathogens, including MRSA, carbapenem-resistant *Klebsiella pneumoniae*, and VRE [95]. These resistant strains are frequently linked to longer hospital stays, higher mortality rates, and increased healthcare expenses. The occurrence of hospital-acquired infections (HAIs) caused by these resistant pathogens is worsened by inadequate infection control measures and the limited availability of new antimicrobial drugs in low- and middle-income countries. In community settings, the easy access to over-the-counter antibiotics and the absence of strict regulations have contributed to the growing resistance among common pathogens like *E. coli* and *Neisseria gonorrhoeae*. Infections from ESBL-producing *E. coli* and multidrug-resistant *Mycobacterium tuberculosis* pose significant challenges in community healthcare, especially in densely populated regions. Additionally, a lack of public awareness about the correct use of antibiotics has exacerbated resistance in outpatient care, making it more difficult to treat common infections [96]. The agricultural sector significantly contributes to the AMR crisis due to the extensive use of antibiotics for growth promotion and disease prevention in livestock and aquaculture. Resistant bacteria from these environments can be transmitted to humans through the food chain, direct contact, and environmental contamination. *Salmonella* spp. and *Campylobacter* spp., often associated with foodborne illnesses, have shown concerning resistance patterns worldwide. Even though there have been attempts to limit antibiotic use in agriculture, especially in wealthier nations, the absence of consistent policies and enforcement in developing areas still hampers efforts to combat AMR [97] (Table 1).

4. Clinical implications of antimicrobial resistance

4.1. Impact on treatment outcomes: mortality, morbidity, and prolonged hospital stays

The emergence of AMR has significantly affected treatment outcomes, resulting in higher rates of mortality, morbidity, and longer hospital stays. Infections caused by resistant organisms, such as multidrug-resistant *Klebsiella pneumoniae*, *Acinetobacter baumannii*, and MRSA, are linked to increased mortality due to delays in effective treatment and a lack of available therapeutic options [103]. Research shows that bloodstream infections from carbapenem-resistant *Enterobacteriaceae* can have mortality rates as high as 50 %, highlighting the serious threat these pathogens pose. Infections from XDR organisms are especially harmful for critically ill and immunocompromised patients,

Table 1

Key settings contributing to antimicrobial resistance: pathogens, drivers, impact, and interventions.

Setting	Common Resistant Pathogens	Drivers of Resistance	Public Health Impact	Proposed Interventions	Refs.
Healthcare	MRSA, carbapenem-resistant <i>K. pneumoniae</i> , VRE, multidrug-resistant <i>Acinetobacter baumannii</i>	Excessive use of broad-spectrum antibiotics, inadequate infection prevention and control (IPC) practices, and insufficient antimicrobial stewardship programs (ASPs)	Increased mortality rates, extended hospital stays, outbreaks in healthcare settings, and rising treatment costs	Implementation of strong ASPs, enhanced IPC measures, creation of quick diagnostic tools, and exploration of alternative therapies	[98]
Community	ESBL-producing <i>E. coli</i> , multidrug-resistant <i>M. tuberculosis</i> , <i>N. gonorrhoeae</i> , macrolide-resistant <i>Streptococcus pneumoniae</i>	The availability of over-the-counter (OTC) antibiotics, along with inappropriate prescribing practices, self-medication, and insufficient public education, are significant concerns	Treatment failures, the spread of resistant infections in outpatient settings, and an increased burden of respiratory and sexually transmitted infections	Public awareness campaigns, tighter regulations on over-the-counter antibiotics, and community-driven stewardship programs	[99]
Agriculture	<i>Salmonella</i> spp., <i>Campylobacter</i> spp., colistin-resistant <i>E. coli</i> , tetracycline-resistant <i>Enterococcus</i>	The use of antibiotics for growth promotion and prophylactic purposes in livestock, combined with inadequate regulatory oversight and the environmental discharge of these antibiotics, raises significant concerns	Outbreaks of foodborne illnesses, transmission from animals to humans, and environmental contamination caused by runoff	Implementing restrictions on antibiotic use in food animals, encouraging the use of vaccines in livestock, and enhancing waste management practices	[100]
Environment	Antibiotic-resistant <i>Pseudomonas</i> spp., <i>Aeromonas</i> spp., resistant environmental <i>E. coli</i>	The release of untreated pharmaceutical waste, agricultural runoff, and inadequate wastewater treatment	The amplification of resistance genes in water systems can lead to contamination of drinking water, which may then spread to both human and animal populations	Enhanced wastewater treatment systems, stricter regulations on the disposal of pharmaceutical waste, and robust environmental monitoring programs	[101]
Veterinary Medicine	Resistant <i>Staphylococcus pseudintermedius</i> , ESBL-producing <i>Enterobacteriaceae</i>	The overuse of antibiotics in pets and the absence of veterinary stewardship programs are concerning issues	The transmission of resistant pathogens to humans poses a significant challenge, especially with the limited therapeutic options available for pets	Creating veterinary antibiotic stewardship programs, formulating guidelines for antibiotic use in pets, and educating the public about associated risks	[102]

who often experience rapid clinical decline. The impact of AMR on morbidity goes beyond the initial infection [104]. Patients frequently suffer from severe complications, including septic shock, organ failure, and secondary infections, which can greatly diminish their quality of

life. For example, community-acquired drug-resistant *E. coli* infections may lead to recurrent urinary tract infections, potentially causing chronic kidney damage. Likewise, resistant respiratory pathogens like multidrug-resistant *Pseudomonas aeruginosa* worsen morbidity in

Table 2

Impact of antimicrobial resistance on treatment outcomes: mortality, morbidity, and healthcare burden with proposed solutions.

Outcome	Examples of Resistant Pathogens	Key Findings	Implications	Proposed Solutions	Refs.
Mortality	CRE, MRSA, XDR <i>A. baumannii</i>	Bloodstream infections caused by CRE can have mortality rates as high as 50 %, while MRSA bloodstream infections can raise mortality by approximately 30 %	There has been a significant increase in loss of life, particularly among vulnerable groups such as ICU patients and those who are immunocompromised	Rapid advancements in the development of new antimicrobials and the establishment of strong infection control measures	[108]
Morbidity	Multidrug-resistant <i>P. aeruginosa</i> , drug-resistant <i>E. coli</i> , VRE	Chronic conditions such as recurrent urinary tract infections, worsening chronic obstructive pulmonary disease, and related complications like organ failure	A decline in quality of life, greater reliance on healthcare systems, and ongoing health issues	Improved public health measures and the promotion of diagnostic stewardship are essential for the early identification of resistant infections	[109]
Prolonged Hospital Stay	XDR <i>A. baumannii</i> , ESBL-producing <i>K. pneumoniae</i>	Infections that are resistant to treatment typically lead to a hospital stay that is 7 to 21 days longer than those caused by infections that are susceptible to treatment	The strain on healthcare resources has led to a reduced availability of hospital beds and an increase in hospital-acquired infections	The implementation of antimicrobial stewardship programs and the use of rapid diagnostic tools are essential to reduce inappropriate antibiotic use	[110]
Economic Burden	Multidrug-resistant organisms (MDROs), CRE	In high-income countries, resistant infections can lead to additional costs of \$700 to \$1500 per patient, while the financial burden is even greater in low-resource settings	Rising healthcare costs are a burden for both individuals and governments, leading to unequal access to care	Global subsidies for new antibiotics, lowering costs through effective management, and encouraging research in resource-limited environments	[111]
Impact on Vulnerable Populations	Drug-resistant <i>Mycobacterium tuberculosis</i> , ESBL-producing <i>E. coli</i>	Infection rates and complications are higher among children, the elderly, and those with weakened immune systems	Worsened outcomes in these groups have led to a greater reliance on last-resort antibiotics	Targeted treatment approaches for high-risk groups, enhanced monitoring in at-risk communities	[85]
Surgical Outcomes	MRSA, VRE	There is an increased risk of surgical site infections (SSIs), which can result in delayed recovery or the need for surgical revisions	Longer recovery times after surgery, higher rates of illness, and complications arising from resistant pathogens	Preoperative screening for resistant pathogens and the use of targeted prophylactic antibiotics	[112]
Global Health Impact	XDR <i>M. tuberculosis</i> , colistin-resistant <i>E. coli</i>	The rise of pan-drug-resistant organisms is particularly concerning in areas where there is significant misuse of antimicrobials and inadequate regulatory measures	The threat to global health security is heightened by the cross-border transmission of resistance	International collaboration for AMR surveillance is essential, as is the need to strengthen regulatory frameworks governing antibiotic use	[113]

individuals with pre-existing conditions such as chronic obstructive pulmonary disease (COPD) and cystic fibrosis [105]. Prolonged hospital stays represent a significant consequence of AMR, necessitated by longer treatment times, constant monitoring, and reliance on last-resort antibiotics. Patients suffering from resistant infections often require more diagnostic tests, specialized care, and isolation, which places a heavy financial strain on healthcare systems. A global study found that individuals with drug-resistant infections spend an average of 7 to 21 days longer in hospitals compared to those with infections that are susceptible to treatment, further taxing already limited healthcare resources. These extended stays not only impact the patients themselves but also diminish the availability of healthcare services for others, worsening systemic issues in infection management [106].

Table 2 illustrates the diverse effects of AMR on treatment outcomes, highlighting increased mortality, morbidity, prolonged hospital stays, and the related economic and healthcare challenges. For example, bloodstream infections caused by carbapenem-resistant *Enterobacteriaceae* (CRE) show alarmingly high mortality rates, reaching up to 50 %, while multidrug-resistant *Pseudomonas aeruginosa* leads to chronic complications in patients with pre-existing conditions. The average hospital stay for those with resistant infections, which ranges from 7 to 21 days, not only burdens healthcare resources but also raises the likelihood of secondary infections acquired in hospitals [107]. Vulnerable groups, such as children, the elderly, and immunocompromised individuals, experience disproportionately severe outcomes, highlighting the unequal effects of AMR. The table also details the financial impact, as resistant infections significantly drive up healthcare costs, especially in low-resource environments. To address these issues, proposed solutions stress the need for antimicrobial stewardship programs, rapid diagnostic tools, preoperative pathogen screening, and global cooperation for AMR monitoring and regulation. This comprehensive strategy aims to provide a focused and effective response to lessen the public health repercussions of AMR (“Antimicrobial resistance”).

4.2. Challenges in diagnosing and treating multidrug-resistant infections

Challenges in diagnosing and treating multidrug-resistant infections are significant for healthcare systems around the globe. The difficulty in identifying resistant pathogens, combined with a limited number of treatment options, complicates the situation. Traditional diagnostic methods, like culture and susceptibility testing, typically take 48 to 72 h to provide results. This delay not only slows down the start of appropriate treatment but also leads to an increased reliance on broad-spectrum antibiotics, which can further promote resistance. In low-resource environments, the lack of access to advanced diagnostic technologies worsens this issue, resulting in many resistant infections going undetected or being misdiagnosed [114]. The absence of rapid and accurate diagnostic tools contributes to the misuse of antimicrobials. For instance, infections caused by multidrug-resistant pathogens such as carbapenem-resistant *Klebsiella pneumoniae* or extensively drug-resistant *Acinetobacter baumannii* may initially be treated with ineffective antibiotics due to the delay in recognizing resistance mechanisms. This mismanagement can result in treatment failures, extended illness, and higher mortality rates [115]. Furthermore, the rise of pathogens with new resistance mechanisms, like plasmid-mediated colistin resistance (mcr genes), adds another layer of complexity to diagnostics, as many standard tests are unable to detect these quickly evolving traits. From a treatment standpoint, the lack of effective antibiotics for multidrug-resistant pathogens presents a serious challenge. Many last-resort antibiotics, including colistin and tigecycline, come with significant toxicity, making their use risky. Additionally, combination therapy, which is often used to address multidrug-resistant infections, lacks strong clinical guidelines and is based on limited evidence, raising the chances of adverse events and treatment failure [27]. The pipeline for new antibiotics is worryingly limited, with only a few agents available to target MDR gram-negative bacteria, leaving healthcare providers

with insufficient options for treating life-threatening infections. In low- and middle-income countries (LMICs), the situation is even more challenging due to a lack of resources, ineffective antimicrobial stewardship programs, and unreliable supply chains for essential antibiotics. The impact of MDR infections in LMICs is disproportionately severe, yet the healthcare systems are often not equipped to handle these cases effectively. Tackling these issues requires a comprehensive strategy that includes investing in rapid diagnostic technologies, creating new antibiotics, and fostering global collaboration to ensure fair access to diagnostics and treatments. Without these efforts, the battle against MDR infections will continue to be a significant and growing crisis [116].

Table 3 outlines the key challenges in diagnosing and treating MDR infections, emphasizing systemic delays, diagnostic shortcomings, and treatment limitations. Delays in diagnosis caused by traditional culture-based methods often lead to inappropriate initial treatments, worsening resistance and increasing patient mortality. Diagnostic shortcomings, such as the failure to identify new resistance mechanisms like mcr-mediated colistin resistance, further complicate the situation. Treatment options for MDR pathogens are limited, with last-resort antibiotics like colistin and tigecycline carrying considerable toxicity risks. Moreover, the absence of standardized guidelines for combination therapies and the rise of PDR organisms heighten the difficulties in treatment [117]. Tackling these challenges necessitates a multifaceted strategy, which includes investing in rapid molecular diagnostics, encouraging the development of new antibiotics, and enhancing global healthcare systems to guarantee fair access and responsible management [118].

4.3. Economic burden of AMR on healthcare systems

The economic burden of AMR on healthcare systems is substantial and growing worldwide. Recent research, including a systematic review and meta-analysis published in PLOS ONE in 2023, has shed light on the financial implications of antibiotic resistance (ABR) in various countries. The findings indicate that patients suffering from resistant infections face higher healthcare expenses due to longer illnesses, extra diagnostic tests, extended hospital stays, and the necessity for more costly medications. The review pointed out that low- and middle-income countries (LMICs) are especially vulnerable, as they often lack the means to effectively address ABR, resulting in increased mortality rates and economic pressure [111].

In the Western Pacific Region, the World Health Organization (WHO) estimates that from 2020 to 2030, AMR could lead to around 5.2 million deaths and incur costs of approximately US\$148 billion due to lost productivity and additional healthcare expenses linked to longer hospital stays. By the end of 2030, patients with antimicrobial-resistant infections in this region are anticipated to spend an extra 172 million days in hospitals. This significant economic burden highlights the urgent need for action to alleviate the effects of AMR on healthcare systems and economies. A recent analysis from the Financial Times suggests that by 2050, global spending on treating drug-resistant infections could reach \$66 billion annually, potentially escalating to \$160 billion. The global workforce and tourism sector are also expected to be adversely affected, with poorer nations facing the most severe economic consequences [124]. Nevertheless, investing \$63 billion each year in new antibiotics and improved treatments for bacterial infections could save 100 million lives, cut health costs by \$97 billion, and boost global GDP by \$960 billion by 2050. This emphasizes the economic advantages of investing in strategies to combat AMR. The financial strain of AMR goes beyond just direct healthcare expenses; it also encompasses wider societal effects, such as decreased productivity from illness and higher mortality rates. A study published in The Lancet in 2024 estimates that antibiotic-resistant bacterial infections could lead to over 39 million deaths globally by 2050, along with an additional 169 million indirect fatalities. The economic repercussions of these deaths could lead to GDP losses between \$1 trillion and \$3.4 trillion by 2030 [125]. These

Table 3
Key challenges in diagnosing and treating multidrug-resistant infections and proposed solutions.

Challenge	Description	Examples	Proposed Solutions	Refs.
Delayed Diagnosis	Traditional diagnostic methods typically require 48 to 72 h, which can delay the start of appropriate treatment	Culture-based identification of carbapenem-resistant and extensively drug-resistant <i>Acinetobacter baumannii</i>	The development and adoption of rapid molecular diagnostics, including PCR and next-generation sequencing, have significantly advanced the field	[119]
Diagnostic Limitations	Standard tests often struggle to identify new resistance mechanisms	Failure to identify <i>mcr</i> -mediated colistin resistance	Improved laboratory capacity, training programs for healthcare professionals, and the incorporation of resistance gene detection	[120]
Limited Therapeutic Options	There are few effective antibiotics available, and the last-resort drugs tend to have high toxicity profiles	The use of colistin or tigecycline for treating multidrug-resistant Gram-negative infections is becoming increasingly important in clinical practice	Encouraging research and development of antibiotics, securing global funding for the creation of new drugs, and conducting clinical trials for combination therapies	[121]
Treatment Failure	There is a lack of guidelines for combination therapies, and we are seeing the emergence of PDR pathogens	PDR <i>Acinetobacter</i> spp. and <i>Pseudomonas</i> spp.	Evidence-based clinical guidelines, studies using real-world data, and collaboration in international research	[122]
Resource Constraints in LMICs	Limited access to advanced diagnostics and the inconsistent availability of essential antibiotics	Inadequate management of MDR tuberculosis in LMICs.	Enhancing healthcare infrastructure, securing international funding for fair access, and advancing antimicrobial stewardship programs	[123]

forecasts highlight the urgent need for comprehensive approaches to tackle AMR, which should include the creation of new antibiotics, enhanced infection control practices, and international cooperation to lessen its economic and health consequences.

5. Public health impact

5.1. AMR as a global health crisis: implications for developing countries and resource-limited settings

AMR is recognized as a global health crisis that has significant implications, particularly for developing countries and areas with limited resources. These regions are confronted with a combination of challenges that worsen the AMR situation, such as limited access to quality healthcare, unregulated use of antibiotics, inadequate infection prevention practices, and a lack of diagnostic tools. The absence of strict antimicrobial stewardship programs contributes to the overuse and misuse of antibiotics, leading to the swift emergence and spread of resistant pathogens. The availability of antibiotics without prescriptions in many low- and middle-income countries (LMICs) further exacerbates the issue [103]. Economic limitations in these resource-constrained settings impede necessary investments in infrastructure to tackle AMR. Many healthcare facilities do not have access to rapid diagnostic tools, which forces clinicians to depend on empirical treatments that may not be suitable when resistant pathogens are present. This not only results in treatment failures but also continues the cycle of resistance. Moreover, the lack of strong surveillance systems means that the true extent of AMR in these areas is often underestimated, hindering effective interventions. To make matters worse, the high cost of newer antibiotics that are effective against multidrug-resistant organisms is often unaffordable for both healthcare systems and individual patients. Developing countries also experience a higher rate of infections linked to poor sanitation, unsafe water, and inadequate healthcare infrastructure, creating an environment conducive to the spread of AMR (Infectious Diseases Society of America [126] et al., 2011). Diseases like tuberculosis (TB) and malaria, which are already widespread in many LMICs, are becoming increasingly resistant to standard treatments, leading to higher rates of illness and death. The emergence of MDR-tuberculosis (MDR-TB) and extensively drug-resistant tuberculosis (XDR-TB) highlights the difficulties in managing AMR in resource-limited areas. Likewise, bloodstream infections from ESBL-producing *Enterobacteriaceae* and carbapenem-resistant *Acinetobacter baumannii* are increasingly prevalent in these regions, often leading to severe consequences [127]. The global ramifications of AMR are particularly alarming for low- and middle-income countries (LMICs) due to their links with global trade, travel, and migration. Pathogens resistant to treatment that originate in

resource-limited settings can spread worldwide, threatening public health on a global scale. Furthermore, the economic impact of AMR is felt more acutely in developing nations, where healthcare costs already take up a large share of household income. Families dealing with AMR-related health issues frequently encounter overwhelming medical expenses, which can trap them in a cycle of poverty and inequality. Tackling AMR in developing countries necessitates a comprehensive strategy that includes bolstering healthcare systems, establishing antimicrobial stewardship initiatives, and investing in surveillance and diagnostic infrastructure. International collaboration and financial assistance are vital to ensure fair access to resources and technologies. Additionally, public health initiatives focused on educating people about proper antibiotic use and enhancing hygiene practices are crucial. Only through unified global efforts can we lessen the disproportionate effects of AMR on resource-limited areas and address its status as a global health emergency [15].

5.2. The role of surveillance systems (e.g., WHO GLASS) in monitoring AMR

Surveillance systems, such as the World Health Organization's Global Antimicrobial Resistance and Use Surveillance System (WHO GLASS), are crucial for monitoring and addressing AMR on a global scale. These systems gather, analyze, and disseminate data on AMR trends, offering vital insights into the prevalence of resistant pathogens in various regions. By standardizing data collection methods and providing a global reporting platform, GLASS helps countries recognize emerging resistance patterns, evaluate the effectiveness of interventions, and shape evidence-based policies [128]. This type of surveillance is key to detecting early signs of resistance to essential antimicrobials, like carbapenems or colistin, which are often the last resort against multidrug-resistant pathogens. Beyond delivering important epidemiological data, GLASS also aids in building capacity in resource-limited settings. It assists countries in establishing national AMR surveillance programs by providing technical guidance, training, and tools to enhance laboratory capabilities. This support is especially vital for low- and middle-income countries (LMICs), where effective AMR monitoring systems are frequently insufficient. By encouraging international collaboration and data sharing, GLASS promotes a coordinated global response to AMR, helping to prioritize research and allocate resources effectively. Nevertheless, the success of these systems relies on ongoing political commitment, sufficient funding, and active participation from all member states to ensure thorough and accurate data collection [129].

The WHO GLASS is essential in the fight against AMR as it standardizes data collection methods, allowing for global comparisons of resistance trends and facilitating early detection of resistance to vital

antibiotics such as carbapenems and colistin. It aids in building capacity in resource-limited areas by providing training, technical tools, and laboratory support, which enhances the AMR surveillance framework (see Table 4). GLASS also supplies evidence-based data that informs national and global AMR policies, promoting international collaboration and data sharing for a unified response. Furthermore, it assists in identifying priority pathogens and regions, ensuring that resources are allocated effectively to address the AMR crisis (“Global Antimicrobial Resistance and Use Surveillance System [[130]S]”).

5.3. AMR’s effect on public health interventions (e.g., vaccination, infection control)

AMR significantly weakens the effectiveness of public health interventions, including vaccination programs and infection control measures, which are essential components of modern healthcare systems. Vaccination, known for its effectiveness in preventing bacterial and viral infections, encounters specific challenges due to AMR. While vaccines help lower the incidence of infections and reduce the reliance on antibiotics, the spread of AMR makes it harder to prevent and treat secondary bacterial infections in those who have been vaccinated. For instance, even with influenza vaccination, secondary infections from resistant bacteria like *Streptococcus pneumoniae* or *Staphylococcus aureus* continue to pose serious risks, especially for vulnerable groups such as the elderly and immunocompromised individuals (“Antimicrobial resistance”). Additionally, AMR undermines the effectiveness of infection control measures in healthcare environments. Standard protocols for managing healthcare-associated infections (HAIs), such as those caused by *Clostridioides difficile* or carbapenem-resistant *Enterobacteriaceae*, often depend on the use of empiric antibiotics. The rise of AMR raises the chances of treatment failures, resulting in longer infections and increased transmission within healthcare facilities. Multidrug-resistant pathogens like MRSA and VRE make it more challenging to control outbreaks, necessitating stricter isolation protocols, improved sterilization practices, and more resource allocation [136].

Infection prevention measures, including hand hygiene, sterilization practices, and environmental decontamination, play a vital role in reducing the impact of AMR. However, AMR complicates and raises the costs of these interventions. For instance, resistant pathogens tend to survive longer on surfaces, which requires more frequent and thorough cleaning. Moreover, the increased reliance on personal protective equipment (PPE) to curb the spread of AMR pathogens adds financial pressure on healthcare systems, particularly in resource-limited areas. Without strong infection control measures, resistant infections can quickly overwhelm healthcare facilities, resulting in higher rates of illness, death, and healthcare-related expenses [137]. Additionally, AMR reduces the indirect benefits of public health measures by limiting the effectiveness of antibiotics used in supportive roles. For example, antibiotics are commonly administered prophylactically to prevent infections during surgeries, cancer treatments, or organ transplants. As resistant pathogens become more prevalent, the likelihood of

post-procedural infections rises, complicating patient recovery and hindering medical progress. This cascading effect of AMR on healthcare interventions underscores the urgent need to incorporate AMR-specific strategies, such as antimicrobial stewardship and the development of new vaccines targeting resistant bacteria, into public health efforts [138].

To tackle the increasing threat of AMR, public health systems need to take a comprehensive approach. This includes expanding vaccination initiatives to prevent infections that are resistant to treatment, enforcing stricter infection control measures, and encouraging responsible use of antimicrobials. Additionally, creating and distributing vaccines that specifically target resistant pathogens, like MRSA or ESBL-producing *E. coli*, could significantly help in alleviating the burdens associated with AMR. By incorporating strategies to combat AMR into public health efforts, healthcare systems can enhance global health protection and maintain the effectiveness of current medical practices [139].

5.4. Real-world applications of molecular research in AMR management: clinical and policy implications

Through molecular research scientists develop and detect AMR through precise treatment methods and the creation of new therapeutic solutions. The main use of molecular diagnostics includes PCR, WGS and CRISPR-based detection platforms that enable quick resistance gene and bacterial strain recognition [140]. The combination of these methods supports clinical choices by helping doctors select antibiotics through genetic resistance information which inhibits overprescribing broad antibiotics while stopping multidrug-resistant (MDR) pathogens from appearing. Through the WHO GLASS surveillance program and other molecular surveillance processes healthcare facilities and public health agencies acquire timely data about pathogen resistance patterns which helps them create specific infection control strategies [141]. Molecular research finds direct therapeutic potential because it enables the establishment of innovative treatment solutions. CRISPR-Cas systems show great potential as a gene editing technique to interrupt resistance genes in bacterial populations thus enabling antibiotics to become effective again. Research shows bacteriophage therapy to be a new alternative to antibiotics by means of viruses that target only resistant bacteria. Medical research has introduced antimicrobial peptides as well as anti-biofilm agents and microbiome-focused therapies that work by interfering with bacterial resistance molecular processes to decrease AMR threats. The newly developed methods demonstrate how molecular medical interventions could transform resistance against antimicrobial drugs during patient treatments [142].

The findings from molecular research enable better policies together with antimicrobial stewardship initiatives that guide medical practices. Molecular epidemiology research provides data needed to create policy decisions that include the restriction of high-risk antibiotics in agricultural and healthcare environments through genetic surveillance data. Hospitals must adopt regulatory frameworks that have rapid molecular diagnostic requirements to decrease use of empirical antibiotics

Table 4
Key contributions of WHO glass in monitoring AMR.

Aspect	Details	Impact	Refs.
Data Collection	Standardized protocols for collecting AMR data across different countries	Facilitates dependable comparisons of resistance patterns on a global scale	[131]
Early Detection	Monitors the resistance levels to essential antimicrobials such as carbapenems and colistin	Recognizes new resistance patterns and guides prompt actions	[132]
Capacity Building	Offers technical support, training, and resources to enhance laboratory capacity in low- and middle-income countries	Enhances AMR surveillance systems in areas with limited resources	[133]
Policy Support	Provides evidence-based data to inform national and global antimicrobial resistance (AMR) policies	Improves decision-making for strategies aimed at containing AMR	[134]
Global Collaboration	Promotes the sharing of data and collaboration among member states	Promotes a unified global effort to address the AMR crisis	[135]
Resource Allocation	Identifies key pathogens and areas that need focused interventions	Makes efficient use of limited resources to effectively tackle AMR	(Infectious Diseases Society of America [126] et al., 2011)

according to policies made by the decision makers [143]. The results of molecular research guide the creation of vaccine programs by providing genetic information that supports development of immunizations against bacteria like *Klebsiella pneumoniae* and *Staphylococcus aureus* that pose high risks.

Multiple government departments need to collaborate to successfully integrate molecular research into resistance management systems. The prevention and prediction of resistance outbreaks depend heavily on fortified networks to share international data and established molecular surveillance systems. The continuous increase of antibiotic-resistant infections demands immediate progress between molecular findings and practical applications to create sustainable containment measures against antibiotic resistance [144].

6. Strategies to combat antimicrobial resistance

6.1. Development of novel antibiotics and alternative therapies

The development of new antibiotics is crucial in addressing the increasing threat of AMR. Once hailed as groundbreaking medical advancements, traditional antibiotics are becoming less effective against MDR pathogens like *Klebsiella pneumoniae*, *Acinetobacter baumannii*, and MRSA. Recent research has concentrated on identifying new classes of antibiotics, enhancing existing medications, and tackling resistance mechanisms such as efflux pumps and enzymatic degradation [145]. For instance, the introduction of β -lactamase inhibitors, such as avibactam, has improved the effectiveness of β -lactam antibiotics against resistant strains. Nevertheless, the high costs, lengthy development processes, and regulatory hurdles contribute to a sluggish pipeline for new antibiotics [146].

Alongside traditional antibiotics, alternative therapies are emerging as viable options to fight AMR. These include bacteriophage therapy, antimicrobial peptides (AMPs), and CRISPR-Cas systems. Bacteriophage therapy uses viruses to specifically target and eliminate bacteria, demonstrating effectiveness against MDR pathogens, especially when antibiotics are ineffective. Likewise, AMPs, which are either naturally occurring or synthetic peptides with broad-spectrum antimicrobial properties, are being studied for their ability to disrupt bacterial

membranes. Additionally, CRISPR-Cas technologies are under investigation for their potential to precisely edit bacterial genomes, directly targeting resistance genes while minimizing off-target effects. Immunotherapies and microbiome-based strategies are also gaining attention as alternative methods. Immunotherapies, such as monoclonal antibodies that target bacterial toxins or virulence factors, assist the host immune system in clearing infections without depending on antibiotics. Microbiome-based therapies, including fecal microbiota transplantation (FMT) and probiotics, aim to restore the balance of gut microbiota, which can be disrupted by infections or antibiotic treatments [147].

The Fig. 3 demonstrates antimicrobial treatment methodologies through its depiction of bacterial cellular elements with corresponding antibiotic restraint mechanisms. Two major antibody types that affect DNA replication and repair include metronidazole along with fluoroquinolones which stop DNA topoisomerase. Using rifampin allows the prevention of mRNA synthesis through its interaction with RNA polymerase. Anti-protein synthesis drugs exert their actions on different ribosomal subunit types where macrolides and streptogramins obstruct the 50S subunit and aminoglycosides and tetracyclines inhibit the 30S subunit [148]. The antimicrobial agents β -lactams and carbapenems dissolve the bacterial cell wall by halting peptidoglycan synthesis thus ending in cell death. Folic acid synthesis inhibitors consisting of sulfonamides along with trimethoprim stop the operation of important metabolic pathways necessary for bacterial survival. The complete strategy demonstrates the wide range of antimicrobial mechanisms that directly support the battle against bacterial infections while tackling the increasing problem of antimicrobial resistance [149].

These therapies have shown potential in preventing infections like *Clostridioides difficile* and reducing the spread of resistance genes. Developing new antibiotics and alternative treatments is crucial, but it demands substantial investment and global cooperation. Public-private partnerships, such as CARB-X and the Global Antibiotic Research and Development Partnership (GARDP), play a vital role in encouraging innovation and tackling market failures in antibiotic development. Furthermore, regulatory frameworks must find a balance between expediting approval processes and ensuring safety and effectiveness. By combining traditional and alternative therapies, healthcare systems can more effectively confront the AMR crisis and protect global health

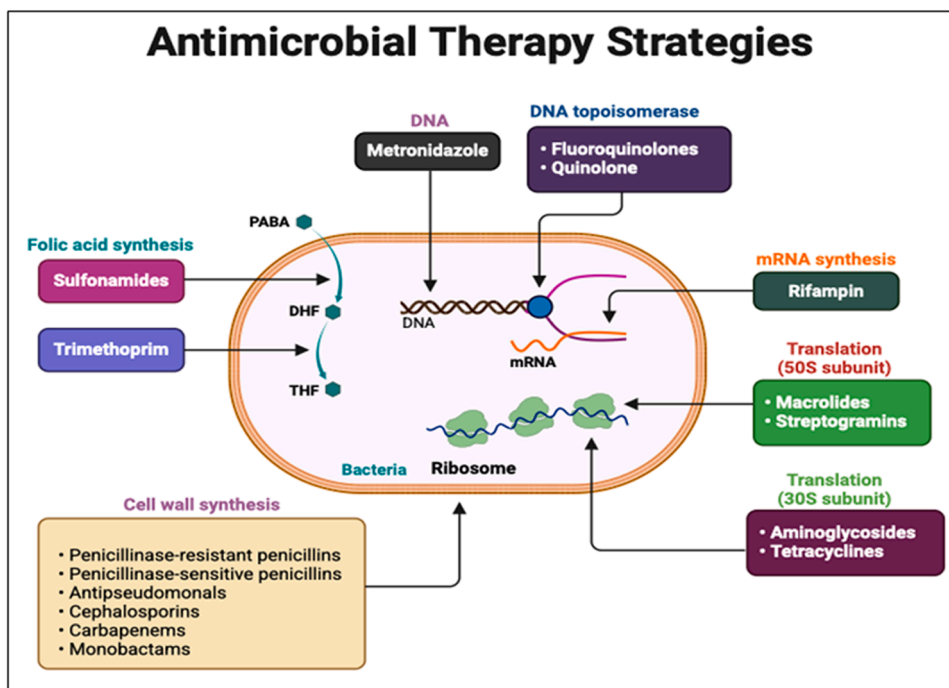


Fig. 3. Mechanisms of action of antimicrobial agents targeting bacterial cellular processes.

[150]. The Table 5 outlines emerging strategies to combat AMR, detailing their mechanisms, benefits, challenges, and future potential. New antibiotics, including β -lactamase inhibitors and lipoglycopeptides, effectively target MDR pathogens but encounter obstacles like high costs and limited approvals. Bacteriophage therapy provides high specificity by using viruses to destroy bacteria, although regulatory challenges remain. Antimicrobial peptides disrupt bacterial membranes with broad-spectrum activity, but issues with stability and toxicity persist. CRISPR-Cas systems hold promise for precise editing of resistance genes, despite delivery difficulties. Immunotherapies, such as monoclonal antibodies, neutralize bacterial toxins, offering targeted solutions but at high production costs. Microbiome-based therapies, including fecal microbiota transplantation, help restore gut microbiota to prevent infections, while nanotechnology-based therapies employ innovative delivery systems to tackle biofilms [151,152]. Adjunctive therapies, like efflux pump inhibitors, enhance existing antibiotics but need further clinical validation. Plant-based compounds, derived from phytochemicals, present cost-effective options with significant potential for antimicrobial discovery.

These strategies show potential for tackling AMR, with continuous research and innovation focused on overcoming existing challenges.

6.2. Antimicrobial stewardship programs: optimizing antibiotic use in clinical and community settings

Antimicrobial stewardship programs (ASPs) are structured initiatives aimed at optimizing antibiotic use in both clinical and community environments to tackle AMR and enhance patient outcomes. In clinical settings, ASPs prioritize evidence-based prescribing, timely reduction of broad-spectrum antibiotics, and dose modifications based on pharmacokinetics and pharmacodynamics (PK/PD) principles [162]. These programs are typically managed by multidisciplinary teams that include infectious disease specialists, pharmacists, and microbiologists, all working together to ensure antibiotics are used rationally and to minimize the rise of multidrug-resistant organisms (MDROs). In community settings, ASPs focus on educating the public about the dangers of antibiotic misuse, raising awareness about AMR, and advocating for restrictions on over-the-counter antibiotics (“[163]).

Key strategies in ASPs involve auditing antibiotic prescriptions, establishing guidelines for empiric therapy, and promoting diagnostic stewardship to cut down on unnecessary antibiotic use. The use of electronic health records (EHRs) allows for real-time tracking of antibiotic consumption and resistance patterns, which supports informed

Table 5
Approaches to combat AMR through novel antibiotics and alternative therapies.

Approach	Examples	Mechanism of Action	Advantages	Challenges	Current Status	Future Potential	Refs.
Novel Antibiotics	β -lactamase inhibitors (e.g., avibactam), lipoglycopeptides (e.g., dalbavancin)	It blocks resistance mechanisms such as β -lactamase activity or the synthesis of the cell wall	Effective against MDR pathogens	High cost, lengthy development timeline	Limited approval of new antibiotic classes	Promising for overcoming resistance in critical priority pathogens	[153]
Bacteriophage Therapy	Phages targeting <i>Pseudomonas aeruginosa</i>	Uses viruses to lyse specific bacteria	Highly specific, minimal off-target effects	Regulatory hurdles, potential resistance	Experimental, some clinical trials ongoing	May be used as an additional therapy, particularly for tailored treatment plans	[154]
Antimicrobial Peptides	LL-37, melittin	Disrupts bacterial membranes	Broad-spectrum activity, low resistance potential	Stability and toxicity concerns	Preclinical and early clinical trials	There is potential to create synthetic derivatives that offer improved stability and effectiveness	[155]
CRISPR-Cas Systems	Gene-editing tools targeting resistance genes	Precisely edits bacterial genomes to disable resistance traits	Highly specific, adaptable for different pathogens	Delivery challenges, ethical considerations	Experimental, mostly in research phases	There is a strong potential for effectively eliminating resistance genes in harmful bacteria	[156]
Immunotherapies	Monoclonal antibodies (e.g., bezlotoxumab for <i>C. difficile</i>)	Neutralizes bacterial toxins or virulence factors	Boosts host immunity, reduces dependence on antibiotics	High production cost, narrow pathogen specificity	FDA approval for select therapies	Expanding applications to address a wide range of resistance mechanisms	[157]
Microbiome-Based Therapies	Fecal microbiota transplantation (FMT), probiotics	Promotes a healthy gut microbiome to help control resistant pathogens	Helps prevent infections and reduces the side effects of antibiotics	Limited data on long-term efficacy	Increasing adoption in clinical practice	The potential of engineered microbiota to provide targeted antimicrobial effects is significant	[158]
Nanotechnology-Based Therapies	Silver nanoparticles, lipid-based nanocarriers	Disrupts bacterial membranes or directly delivers antibiotics to pathogens	Improved drug delivery with the capability to penetrate biofilms	Safety issues and the ability to scale for large-scale production	Preclinical and early clinical trials	There is a strong potential for creating next-generation antimicrobials that can be delivered with precision	[159]
Adjunctive Therapies	Antibiotic adjuvants (e.g., efflux pump inhibitors)	Inhibits bacterial resistance mechanisms, improving the effectiveness of antibiotics	Enhances the effectiveness of current antibiotics	Limited success in clinical trials	Research is ongoing, and there are several adjuvants currently in development	May help prolong the effectiveness of current antibiotic classes	[160]
Plant-Based Compounds	Phytochemicals (e.g., flavonoids, alkaloids)	Disrupts the metabolism of bacteria or targets factors that contribute to their virulence	Affordable production capabilities with natural resources	Variability in potency, challenges in standardization	Early-stage research	Promising sources for new antimicrobial scaffolds	[161]

decision-making. Despite their proven effectiveness, ASPs encounter challenges such as limited resources, a shortage of trained staff, and pushback from prescribers. Nevertheless, the growth of telemedicine and the creation of region-specific stewardship guidelines can significantly boost their effectiveness. Successful ASPs have shown reductions in antibiotic use, fewer hospital-acquired infections, and improved cost-effectiveness [164].

Looking ahead, ASPs aim to utilize artificial intelligence (AI) for tailored antibiotic recommendations, develop community-level stewardship frameworks, and incorporate ASPs into national action plans against AMR. It is essential to strengthen ASPs in low- and middle-income countries through better resource allocation, capacity building, and global collaboration to effectively address the worldwide challenge of AMR [165].

This Table 6 offers a detailed look at antimicrobial stewardship programs (ASPs) in both clinical and community environments, highlighting their goals, strategies, challenges, and future possibilities. In clinical settings, ASPs focus on optimizing antibiotic use through prescription audits, evidence-based guidelines, and training for healthcare providers. On the community side, ASPs prioritize public education, awareness initiatives, and regulations limiting over-the-counter antibiotic sales. Essential tools like electronic health records in hospitals and public health campaigns in communities are vital for their success. Monitoring methods, such as tracking antibiotic usage and conducting surveys, are important for evaluating impact and guiding improvements. Although there are challenges like funding constraints and cultural obstacles, ASPs hold great potential for reducing AMR by promoting responsible antibiotic use and increasing public awareness. Future developments, including AI-driven solutions, global partnerships, and integrated policies, could further enhance the reach and effectiveness of ASPs around the world (Worldwide Antimicrobial Resistance National/International Network Group [[166]ING] Collaborators, 2023).

6.3. The role of rapid diagnostics in curbing resistance

The rise of AMR presents a significant challenge to global public health, necessitating innovative strategies to curb its spread. Rapid diagnostic technologies are crucial in tackling this issue by allowing for the swift and accurate identification of pathogens and their resistance profiles. Unlike traditional methods that can take a long time, rapid diagnostics deliver results within hours, enabling early and targeted treatment [103]. This approach helps reduce the unnecessary use of broad-spectrum antibiotics, which is a major factor in the development of resistant strains, ensuring that patients receive the most effective therapy without delay. One of the key benefits of rapid diagnostics is their capability to distinguish between bacterial, viral, and fungal

infections. For example, PCR assays and next-generation sequencing (NGS) techniques can accurately identify pathogens and resistance genes. Point-of-care (POC) tests, like lateral flow immunoassays, allow for testing at the bedside, making them especially useful in resource-limited settings [143]. These advancements not only enhance patient outcomes but also help prevent the misuse of antibiotics in non-bacterial infections. By promoting appropriate antimicrobial use, rapid diagnostics play a vital role in stewardship programs aimed at combating AMR.

However, despite their advantages, the widespread implementation of rapid diagnostics encounters several challenges. High costs, inadequate infrastructure, and the requirement for technical expertise limit their availability, particularly in low- and middle-income countries (LMICs). Furthermore, false positives or negatives in some rapid tests can result in incorrect treatment choices. Addressing these obstacles necessitates coordinated efforts, including investments in affordable technologies, training initiatives for healthcare professionals, and the incorporation of diagnostics into national AMR action plans. Collaborative efforts among governments, industries, and academic institutions can further foster innovation and scalability [178]. The future of rapid diagnostics is incredibly promising, especially with the incorporation of artificial intelligence (AI) and machine learning (ML) algorithms. These advancements can improve how we interpret diagnostic data, allowing for predictive insights and immediate decision-making. Portable diagnostic devices and progress in microfluidics are set to change the landscape of testing in remote and underserved areas. By connecting rapid diagnostics with antimicrobial stewardship, these innovations could significantly impact the global fight against AMR. As diagnostic technologies advance, their successful application will be crucial in addressing resistance challenges [179].

7. Conclusion

To tackle AMR successfully the world requires coordinated efforts alongside equal distribution of resources then combining molecular research with public health intervention strategies. Current research focuses exclusively on molecular or epidemiological aspects but this review combines these elements to deliver an entire view of AMR development and handling methods. The review emphasizes the value of advanced molecular surveillance programs (WHO GLASS) while discussing newly available treatment methods (CRISPR-Cas systems and phage therapy) together with diagnostics operated by AI which former reviews did not address. This review assesses important deficiencies in current AMR management strategies which the author recommends solving through multilateral methods that combine molecular analysis with policy implementation and fresh treatment solutions. The study

Table 6
Comprehensive aspects of antimicrobial stewardship programs.

Aspect	Clinical Settings	Community Settings	Refs.
Objective	Enhance the use of antibiotics to achieve better health outcomes and minimize AMR	Prevent the misuse of antibiotics and increase public awareness	[167]
Key Strategies	Prescribing based on evidence, optimizing doses, and de-escalating treatment	Public education, limiting over-the-counter antibiotics, awareness initiatives	[168]
Team Composition	Infectious disease specialists, pharmacists, and microbiologists	Healthcare providers, pharmacists, policymakers	[169]
Tools and Resources	EHRs, diagnostic stewardship, resistance surveillance	Public health campaigns, telemedicine platforms	[170]
Monitoring Mechanisms	Prescription audits, antibiotic usage metrics, infection control audits	Surveys on antibiotic knowledge and practices, community-level antibiotic usage data	[171]
Educational Interventions	Training for healthcare workers, workshops, and clinical guidelines	Public campaigns, school programs, and community workshops	[172]
Challenges	Limited resources, prescriber resistance, lack of training	Misconceptions about antibiotics, lack of awareness	[173]
Barriers in Implementation	Inadequate funding, infrastructure constraints, cultural barriers	Accessibility issues in rural areas, resistance to behavior change	[174]
Impact	Reduced MDR infections, optimized antibiotic use, cost savings	Increased public knowledge, reduced misuse	[175]
Future Directions	AI-driven recommendations, region-specific frameworks	Community-level stewardship integration, global action plans	[176]
Policy Support	National AMR policies, hospital accreditation requirements	Legislation to restrict OTC antibiotic sales, public health policies	[“B4691.pdf”] [177]
Global Collaboration	Sharing resistance data, adopting WHO ASP frameworks	International campaigns to improve public awareness of AMR	[177]

creates a research and policy development foundation by showing how to connect genetic AMR factors with associated impacts on healthcare and costs for future efforts in global AMR containment.

This review combines extensive research between AMR biological mechanisms with epidemiological statistics and contemporary treatment developments thus establishing itself unlike standard literature that studies individual elements. The review takes an expansive approach by linking three aspects typically presented separately to present a thorough view of AMR development and public health impacts. Real-time AMR monitoring and intervention preparation requires modern molecular surveillance equipment such as whole-genome sequencing combined with artificial intelligence diagnostics systems. The main advancement of this paper lies in examining untraditional therapeutic approaches that combine CRISPR-based gene editing with bacteriophage therapy along with antimicrobial peptides and microbiome-targeted interventions yet conventional AMR research lacks this attention. The review highlights how resistance mechanisms combine with global public health challenges so the authors push for combined molecular research with policy-driven interventions. This review unites AMR genetic factors to their health, economic and social implications to establish a strategic basis for conducting future research and designing antimicrobial stewardship and worldwide policy measures to fight the rising AMR threat.

CRediT authorship contribution statement

Ghazala Muteeb: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Raisa Nazir Ahmed Kazi:** Writing – original draft, Visualization, Methodology, Investigation. **Mohammad Aatif:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Asim Azhar:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Mohamed El Oirdi:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Mohd Farhan:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Deanship of Scientific Research, Vice Presidency for Graduate Studies and Scientific Research, King Faisal University, Saudi Arabia [Grant No.: KF251377].

References

- [1] Solanki S, Kumar DH. Antimicrobial resistance: molecular drivers and underlying mechanisms. *J Med Surg Public Health* 2024;3:100122. <https://doi.org/10.1016/j.glmedi.2024.100122>.
- [2] Muteeb G. Network meta-analysis of antibiotic resistance patterns in gram-negative bacterial infections: a comparative study of carbapenems, fluoroquinolones, and aminoglycosides. *Front Microbiol* 2023;14:1304011. <https://doi.org/10.3389/fmicb.2023.1304011>.
- [3] Sawa T, Kooguchi K, Moriyama K. Molecular diversity of extended-spectrum β -lactamases and carbapenemases, and antimicrobial resistance. *J Intensive Care* 2020;8:13. <https://doi.org/10.1186/s40560-020-0429-6>.
- [4] Zhuang M, Yan W, Xiong Y, Wu Z, Cao Y, Sanganyado E, Siame BA, Chen L, Kashi Y, Leung KY. Horizontal plasmid transfer promotes antibiotic resistance in selected bacteria in Chinese frog farms. *Environ Int* 2024;190:108905. <https://doi.org/10.1016/j.envint.2024.108905>.
- [5] Gaurav A, Bakht P, Saini M, Pandey S, Pathania R. Role of bacterial efflux pumps in antibiotic resistance, virulence, and strategies to discover novel efflux pump inhibitors. *Microbiology (Reading, Engl.)* 2023;169:001333. <https://doi.org/10.1099/mic.0.001333>.
- [6] Li X-Z, Nikaido H. Efflux-mediated drug resistance in bacteria: an update. *Drugs* 2009;69:1555–623. <https://doi.org/10.2165/11317030-000000000-00000>.
- [7] Sethuvel DPM, Bakthavatchalam YD, Karthik M, Irulappan M, Shrivastava R, Periasamy H, Veeraraghavan B. β -lactam resistance in ESKAPE pathogens mediated through modifications in penicillin-binding proteins: an overview. *Infect Dis Ther* 2023;12:829–41. <https://doi.org/10.1007/s40121-023-00771-8>.
- [8] Zhu S, Yang B, Wang Z, Liu Y. Augmented dissemination of antibiotic resistance elicited by non-antibiotic factors. *Ecotoxicol Environ Saf* 2023;262:115124. <https://doi.org/10.1016/j.ecoenv.2023.115124>.
- [9] Noel HR, Petrey JR, Palmer LD. Mobile genetic elements in *Acinetobacter* antibiotic-resistance acquisition and dissemination. *Ann N Y Acad Sci* 2022;1518:166–82. <https://doi.org/10.1111/nyas.14918>.
- [10] NIHR Global Health Research Unit on Genomic Surveillance of AMR. Whole-genome sequencing as part of national and international surveillance programmes for antimicrobial resistance: a roadmap. *BMJ Glob Health* 2020;5:e002244. <https://doi.org/10.1136/bmjgh-2019-002244>.
- [11] Sheu C-C, Chang Y-T, Lin S-Y, Chen Y-H, Hsueh P-R. Infections caused by carbapenem-resistant Enterobacteriaceae: an update on therapeutic options. *Front Microbiol* 2019;10:80. <https://doi.org/10.3389/fmicb.2019.00080>.
- [12] Kunz Coyne AJ, El Ghali A, Holger D, Rebold N, Rybak MJ. Therapeutic strategies for emerging multidrug-resistant *Pseudomonas aeruginosa*. *Infect Dis Ther* 2022;11:661–82. <https://doi.org/10.1007/s40121-022-00591-2>.
- [13] Al-Fadhli AH, Jamal WY. Recent advances in gene-editing approaches for tackling antibiotic resistance threats: a review. *Front Cell Infect Microbiol* 2024;14:1410115. <https://doi.org/10.3389/fcimb.2024.1410115>.
- [14] Oliveira M, Antunes W, Mota S, Madureira-Carvalho A, Dinis-Oliveira RJ, Dias DSD. An overview of the recent advances in antimicrobial resistance. *Microorganisms* 2024;12:1920. <https://doi.org/10.3390/microorganisms12091920>.
- [15] Coque TM, Cantón R, Pérez-Cobas AE, Fernández-de-Bobadilla MD, Baquero F. Antimicrobial resistance in the Global health network: known unknowns and challenges for efficient responses in the 21st century. *Microorganisms* 2023;11:1050. <https://doi.org/10.3390/microorganisms11041050>.
- [16] Varela MF, Stephen J, Lekshmi M, Ojha M, Wenzel N, Sanford LM, Hernandez AJ, Parvathi A, Kumar SH. Bacterial resistance to antimicrobial agents. *Antibiotics* 2021;10:593. <https://doi.org/10.3390/antibiotics10050593>.
- [17] Sharma S, Chauhan A, Ranjan A, Mathkor DM, Haque S, Ramniwas S, Tuli HS, Jindal T, Yadav V. Emerging challenges in antimicrobial resistance: implications for pathogenic microorganisms, novel antibiotics, and their impact on sustainability. *Front Microbiol* 2024;15:1403168. <https://doi.org/10.3389/fmicb.2024.1403168>.
- [18] Hussain HI, Aqib AI, Seleem MN, Shabbir MA, Hao H, Iqbal Z, Kulyar MF-E-A, Zaheer T, Li K. Genetic basis of molecular mechanisms in β -lactam resistant gram-negative bacteria. *Microb Pathog* 2021;158:105040. <https://doi.org/10.1016/j.micpath.2021.105040>.
- [19] Boyd SE, Livermore DM, Hooper DC, Hope WW. Metallo- β -lactamases: structure, function, epidemiology, treatment options, and the development pipeline. *Antimicrob Agents Chemother* 2020;64:e00397. <https://doi.org/10.1128/AAC.00397-20>.
- [20] Lorusso AB, Carrara JA, Barroso CDN, Tuon FF, Faoro H. Role of efflux pumps on antimicrobial resistance in *Pseudomonas aeruginosa*. *Int J Mol Sci* 2022;23:15779. <https://doi.org/10.3390/ijms232415779>.
- [21] Li X-Z, Plésiat P, Nikaido H. The challenge of efflux-mediated antibiotic resistance in gram-negative bacteria. *Clin Microbiol Rev* 2015;28:337–418. <https://doi.org/10.1128/CMR.00117-14>.
- [22] Wielders CLC, Fluit AC, Brisse S, Verhoef J, Schmitz FJ. *mecA* gene is widely disseminated in *Staphylococcus aureus* population. *J Clin Microbiol* 2002;40:3970–5. <https://doi.org/10.1128/JCM.40.11.3970-3975.2002>.
- [23] Minarini LAR, Darini ALC. Mutations in the quinolone resistance-determining regions of *gyrA* and *parC* in Enterobacteriaceae isolates from Brazil. *Braz J Microbiol* 2012;43:1309–14. <https://doi.org/10.1590/S1517-838220120004000010> [Publication of the Brazilian Society for Microbiology].
- [24] Partridge SR, Kwong SM, Firth N, Jensen SO. Mobile genetic elements associated with antimicrobial resistance. *Clin Microbiol Rev* 2018;31:e00088. <https://doi.org/10.1128/CMR.00088-17>.
- [25] Ali N, Ali I, Din AU, Akhtar K, He B, Wen R. Integrins in the age of antibiotic resistance: evolution, mechanisms, and environmental implications: a review. *Microorganisms* 2024;12:2579. <https://doi.org/10.3390/microorganisms12122579>.
- [26] Newton-Foot M, Snyman Y, Maloba MRB, Whitelaw AC. Plasmid-mediated *mcr-1* colistin resistance in *Escherichia coli* and *Klebsiella* spp. clinical isolates from the Western Cape region of South Africa. *Antimicrob Resist Infect Control* 2017;6:78. <https://doi.org/10.1186/s13756-017-0234-8>.
- [27] Mondal AH, Khare K, Saxena P, Debnath P, Mukhopadhyay K, Yadav D. A review on Colistin resistance: an antibiotic of last resort. *Microorganisms* 2024;12:772. <https://doi.org/10.3390/microorganisms12040772>.
- [28] Trauner A, Banaei-Esfahani A, Gygli SM, Warmer P, Feldmann J, Zampieri M, Borrell S, Collins BC, Beisel C, Aebersold R, Gagneux S. Expression dysregulation as a mediator of fitness costs in antibiotic resistance. *Antimicrob Agents Chemother* 2021;65:e0050421. <https://doi.org/10.1128/AAC.00504-21>.
- [29] Liu CSC, Pandey R. Integrative genomics would strengthen AMR understanding through ONE health approach. *Heliyon* 2024;10:e34719. <https://doi.org/10.1016/j.heliyon.2024.e34719>.
- [30] Jian Z, Zeng L, Xu T, Sun S, Yan S, Yang L, Huang Y, Jia J, Dou T. Antibiotic resistance genes in bacteria: occurrence, spread, and control. *J Basic Microbiol* 2021;61:1049–70. <https://doi.org/10.1002/jobm.202100201>.

- [31] Sánchez-Céspedes J, Sáez-López E, Frimodt-Møller N, Vila J, Soto SM. Effects of a mutation in the *gyrA* gene on the virulence of uropathogenic *Escherichia coli*. *Antimicrob Agents Chemother* 2015;59:4662–8. <https://doi.org/10.1128/AAC.00665-15>.
- [32] Sachan RSK, Mistry V, Dholaria M, Rana A, Devgon I, Ali I, Iqbal J, Eldin SM, Mohammad SATAR, Bawazeer S, Dutta J, Karnwal A. Overcoming *mycobacterium tuberculosis* drug resistance: novel medications and repositioning strategies. *ACS Omega* 2023;8:32244–57. <https://doi.org/10.1021/acsomega.3c02563>.
- [33] Pfeifer E, Bonnin RA, Rocha EPC. Phage-plasmids spread antibiotic resistance genes through infection and lysogenic conversion. *mBio* 2022;13:e0185122. <https://doi.org/10.1128/mbio.01851-22>.
- [34] Muteeb G, Rehman MT, Shahwan M, Aatif M. Origin of antibiotics and antibiotic resistance, and their impacts on drug development: a narrative review. *Pharmaceuticals* 2023;16:1615. <https://doi.org/10.3390/ph16111615>.
- [35] Vrancianu CO, Popa LI, Bleotu C, Chifiriuc MC. Targeting plasmids to limit acquisition and transmission of antimicrobial resistance. *Front Microbiol* 2020;11:761. <https://doi.org/10.3389/fmicb.2020.00761>.
- [36] Cuzon G, Naas T, Nordmann P. Functional characterization of Tn4401, a Tn3-based transposon involved in blaKPC gene mobilization. *Antimicrob Agents Chemother* 2011;55:5370–3. <https://doi.org/10.1128/AAC.05202-11>.
- [37] Ligozzi M, Lo CG, Fontana R. vanA gene cluster in a vancomycin-resistant clinical isolate of *Bacillus circulans*. *Antimicrob Agents Chemother* 1998;42:2055–9. <https://doi.org/10.1128/AAC.42.8.2055>.
- [38] Schrader SM, Vaubourgeix J, Nathan C. Biology of antimicrobial resistance and approaches to combat it. *Sci Transl Med* 2020;12:eaa6992. <https://doi.org/10.1126/scitranslmed.aaz6992>.
- [39] Sharma A, Gupta VK, Pathania R. Efflux pump inhibitors for bacterial pathogens: from bench to bedside. *Indian J Med Res* 2019;149:129–45. <https://doi.org/10.4103/ijmr.IJMR.2079.17>.
- [40] López CA, Travers T, Pos KM, Zgurskaya HI, Gnanakaran S. Dynamics of intact MexAB-OprM efflux pump: focusing on the MexA-OprM interface. *Sci Rep* 2017;7:16521. <https://doi.org/10.1038/s41598-017-16497-w>.
- [41] Dauri L, Orange F, Taveau J-C, Verchère A, Monlezun L, Gounou C, Marredy RKR, Picard M, Broutin I, Pos KM, Lambert O. Tripartite assembly of RND multidrug efflux pumps. *Nat Commun* 2016;7:10731. <https://doi.org/10.1038/ncomms10731>.
- [42] Koo SH. Overexpression of efflux pump in multidrug-resistant *Pseudomonas aeruginosa*: how you will discover and treat it? *Infect Chemother* 2015;47:142–4. <https://doi.org/10.3947/ic.2015.47.2.142>.
- [43] Mora-Ochomogo M, Lohans CT. β -lactam antibiotic targets and resistance mechanisms: from covalent inhibitors to substrates. *RSC Med Chem* 2021;12:1623–39. <https://doi.org/10.1039/d1md00200g>.
- [44] Macheboeuf P, Contreras-Martel C, Job V, Dideberg O, Dessen A. Penicillin binding proteins: key players in bacterial cell cycle and drug resistance processes. *FEMS Microbiol Rev* 2006;30:673–91. <https://doi.org/10.1111/j.1574-6976.2006.00024.x>.
- [45] Castanheira M, Simmer PJ, Bradford PA. Extended-spectrum β -lactamases: an update on their characteristics, epidemiology and detection. *JAC Antimicrob Resist* 2021;3:dlab092. <https://doi.org/10.1093/jacamr/dlab092>.
- [46] Baran A, Kwiatkowska A, Potocki L. Antibiotics and bacterial resistance-A short story of an endless arms race. *Int J Mol Sci* 2023;24:5777. <https://doi.org/10.3390/ijms24065777>.
- [47] Fergestad ME, Stamsås GA, Morales AD, Salehian Z, Westeson Y, Kjos M. Penicillin-binding protein PBP2a provides variable levels of protection toward different β -lactams in *Staphylococcus aureus* RN4220. *Microbiologyopen* 2020;9:e1057. <https://doi.org/10.1002/mbo3.1057>.
- [48] Hooper DC, Jacoby GA. Mechanisms of drug resistance: quinolone resistance. *Ann N Y Acad Sci* 2015;1354:12–31. <https://doi.org/10.1111/nyas.12830>.
- [49] De Gaetano GV, Lentini G, Famà A, Coppolino F, Beninati C. Antimicrobial resistance: two-component regulatory systems and multidrug efflux pumps. *Antibiotics* 2023;12:965. <https://doi.org/10.3390/antibiotics12060965>.
- [50] Reygaert WC. An overview of the antimicrobial resistance mechanisms of bacteria. *AIMS Microbiol* 2018;4:482–501. <https://doi.org/10.3934/microbiol.2018.3.482>.
- [51] Zhou G, Wang Q, Wang Y, Wen X, Peng H, Peng R, Shi Q, Xie X, Li L. Outer membrane porins contribute to antimicrobial resistance in gram-negative bacteria. *Microorganisms* 2023;11:1690. <https://doi.org/10.3390/microorganisms11071690>.
- [52] Sugawara E, Kojima S, Nikaido H. *Klebsiella pneumoniae* major porins OmpK35 and OmpK36 allow more efficient diffusion of β -lactams than their *Escherichia coli* homologs OmpF and OmpC. *J Bacteriol* 2016;198:3200–8. <https://doi.org/10.1128/JB.00590-16>.
- [53] Gellatly SL, Needham B, Madera L, Trent MS, Hancock REW. The *Pseudomonas aeruginosa* PhoP-PhoQ two-component regulatory system is induced upon interaction with epithelial cells and controls cytotoxicity and inflammation. *Infect Immun* 2012;80:3122–31. <https://doi.org/10.1128/IAI.00382-12>.
- [54] Fernández L, Hancock REW. Adaptive and mutational resistance: role of porins and efflux pumps in drug resistance. *Clin Microbiol Rev* 2012;25:661–81. <https://doi.org/10.1128/CMR.00043-12>.
- [55] Zhao Y, Xu H, Wang H, Wang P, Chen S. Multidrug resistance in *Pseudomonas aeruginosa*: genetic control mechanisms and therapeutic advances. *Mol Biomed* 2024;5:62. <https://doi.org/10.1186/s43556-024-00221-y>.
- [56] Tajer L, Paillart J-C, Dib H, Sabatier J-M, Fajloun Z, Abi KZ. Molecular mechanisms of bacterial resistance to antimicrobial peptides in the modern era: an updated review. *Microorganisms* 2024;12:1259. <https://doi.org/10.3390/microorganisms12071259>.
- [57] Liu Y-Y, Wang Y, Walsh TR, Yi LX, Zhang R, Spencer J, Doi Y, Tian G, Dong B, Huang X, Yu L-F, Gu D, Ren H, Chen X, Lv L, He D, Zhou H, Liang Z, Liu J-H, Shen J. Emergence of plasmid-mediated colistin resistance mechanism MCR-1 in animals and human beings in China: a microbiological and molecular biological study. *Lancet Infect Dis* 2016;16:161–8. [https://doi.org/10.1016/S1473-3099\(15\)00424-7](https://doi.org/10.1016/S1473-3099(15)00424-7).
- [58] Petrosillo N, Taglietti F, Granata G. Treatment options for Colistin resistant *Klebsiella pneumoniae*: present and future. *J Clin Med* 2019;8:934. <https://doi.org/10.3390/jcm8070934>.
- [59] Karade S, Sen S, Shergill SPS, Jani K, Shouche Y, Gupta RM. Whole genome sequence of colistin-resistant *Escherichia coli* from western India. *Med J Armed Forces India* 2021;77:297–301. <https://doi.org/10.1016/j.mjafi.2020.10.008>.
- [60] Sharma S, Mohler J, Mahajan SD, Schwartz SA, Bruggemann L, Aalinkel R. Microbial biofilm: a review on formation, infection, antibiotic resistance, control measures, and innovative treatment. *Microorganisms* 2023;11:1614. <https://doi.org/10.3390/microorganisms11061614>.
- [61] Schlichter KY, Muthuraman S, Nisaa K, Ben-Zvi A, Karsagi BDL, Shagan M, Brandis A, Mehlman T, Gopas J, Saravana KR, Kushmaro A. *Pseudomonas aeruginosa* quorum sensing and biofilm attenuation by a di-hydroxy derivative of piperlongumine (PL-18). *Biofilm* 2024;8:100215. <https://doi.org/10.1016/j.biofilm.2024.100215>.
- [62] Van Acker H, Coenye T. The role of efflux and physiological adaptation in biofilm tolerance and resistance. *J Biol Chem* 2016;291:12565–72. <https://doi.org/10.1074/jbc.R115.707257>.
- [63] Dutt Y, Dhiman R, Singh T, Vibhuti A, Gupta A, Pandey RP, Raj VS, Chang C-M, Priyadarshini A. The association between biofilm formation and antimicrobial resistance with possible ingenious bio-remedial approaches. *Antibiotics (Basel)* 2022;11:930. <https://doi.org/10.3390/antibiotics11070930>.
- [64] Kunnath AP, Suodha SM, Chellappan DK, Chellian J, Palaniveloo K. Bacterial persister cells and development of antibiotic resistance in chronic infections: an update. *Br J Biomed Sci* 2024;81:12958. <https://doi.org/10.3389/bjbs.2024.12958>.
- [65] Singh G, Yadav M, Ghosh C, Rathore JS. Bacterial toxin-antitoxin modules: classification, functions, and association with persistence. *Curr Res Microb Sci* 2021;2:100047. <https://doi.org/10.1016/j.crmicr.2021.100047>.
- [66] Wood TK, Knebel SJ, Kwan BW. Bacterial persister cell formation and dormancy. *Appl Environ Microbiol* 2013;79:7116–21. <https://doi.org/10.1128/AEM.02636-13>.
- [67] Burmeister AR. Horizontal gene transfer. *Evol Med Public Health* 2015;2015:193–4. <https://doi.org/10.1093/emph/eov018>.
- [68] Virolle C, Goldlust K, Djermoun S, Bigot S, Lesterlin C. Plasmid transfer by conjugation in gram-negative bacteria: from the cellular to the community level. *Genes (Basel)* 2020;11:1239. <https://doi.org/10.3390/genes11111239>.
- [69] Varani A, He S, Siguier P, Ross K, Chandler M. The IS6 family, a clinically important group of insertion sequences including IS26. *Mob DNA* 2021;12:11. <https://doi.org/10.1186/s13100-021-00239-x>.
- [70] Seow VY, Tsygelnyska O, Biais N. Multisite transformation in *Neisseria gonorrhoeae*: insights on transformations mechanisms and new genetic modification protocols. *Front Microbiol* 2023;14:1178128. <https://doi.org/10.3389/fmicb.2023.1178128>.
- [71] Heap JT, Ehsaan M, Cooksley CM, Ng Y-K, Cartman ST, Winzer K, Minton NP. Integration of DNA into bacterial chromosomes from plasmids without a counter-selection marker. *Nucleic Acids Res* 2012;40:e59. <https://doi.org/10.1093/nar/okr1321>.
- [72] Chiang YN, Penadés JR, Chen J. Genetic transduction by phages and chromosomal islands: the new and noncanonical. *PLoS Pathog* 2019;15:e1007878. <https://doi.org/10.1371/journal.ppat.1007878>.
- [73] Fillol-Salom A, Alsaadi A, Sousa JAM de, Zhong L, Foster KR, Rocha EPC, Penadés JR, Ingmer H, Haaber J. Bacteriophages benefit from generalized transduction. *PLoS Pathog* 2019;15:e1007888. <https://doi.org/10.1371/journal.ppat.1007888>.
- [74] Bhat BA, Mir RA, Qadri H, Dhiman R, Almilaibary A, Alkhanani M, Mir MA. Integrations in the development of antimicrobial resistance: critical review and perspectives. *Front Microbiol* 2023;14:1231938. <https://doi.org/10.3389/fmicb.2023.1231938>.
- [75] Lermiaux NA, Cameron ADS. Horizontal transfer of antibiotic resistance genes in clinical environments. *Can J Microbiol* 2019;65:34–44. <https://doi.org/10.1139/cjm-2018-0275>.
- [76] Taggar G, Attiq RM, Boerlin P, Diarra MS. Molecular epidemiology of carbapenemases in enterobacteriales from humans, animals, food and the environment. *Antibiotics (Basel)* 2020;9:693. <https://doi.org/10.3390/antibiotics9100693>.
- [77] Mejía-Limones I, Andrade-Molina D, Morey-León G, Hidalgo-Olmedo JC, Chang-Asinc JG, Fernández-Cadena JC, Rojas M. Whole-genome sequencing of *Klebsiella pneumoniae* MDR circulating in a pediatric hospital setting: a comprehensive genome analysis of isolates from Guayaquil, Ecuador. *BMC Genom* 2024;25:928. <https://doi.org/10.1186/s12864-024-10835-9>.
- [78] De Oliveira Santos JV, Da Costa Júnior SD, De Fátima Ramos Dos Santos Medeiros SM, Cavalcanti IDL, De Souza JB, Coriolano DL, Da Silva WRC, Alves MHME, Cavalcanti IMF. Panorama of bacterial infections caused by epidemic resistant strains. *Curr Microbiol* 2022;79:175. <https://doi.org/10.1007/s00284-022-02875-9>.
- [79] Bhownik D, Chetri S, Das B.J., Dhar C.D, Bhattacharjee A.. 2021. Distribution of virulence genes and SCCmec types among methicillin-resistant *Staphylococcus aureus* of clinical and environmental origin: a study from community of Assam, India. *BMC Res Notes* 14:58. doi:10.1186/s13104-021-05473-3.

- [80] Kariuki S, Kering K, Wairimu C, Onsare R, Mbae C. Antimicrobial resistance rates and surveillance in Sub-Saharan Africa: where are we now? *Infect Drug Resist* 2022;15:3589–609. <https://doi.org/10.2147/IDR.S342753>.
- [81] Huang J, Lan F, Lu Y, Li B. Characterization of integrons and antimicrobial resistance in *Escherichia coli* sequence type 131 isolates. *Can J Infect Dis Med Microbiol* 2020;2020:3826186. <https://doi.org/10.1155/2020/3826186>.
- [82] Nieto-Saucedo J.R., López-Jacome L.E., Franco-Cendejas R., Colín-Castro C.A., Hernández-Duran M., Rivera-Garay L.R., Zamarripa-Martínez K.S., Mosqueda-Gómez J.L. 2023. Carbapenem-resistant gram-negative Bacilli characterization in a tertiary care center from El Bajío, Mexico. *Antibiotics (Basel)* 12:1295. [doi:10.3390/antibiotics12081295](https://doi.org/10.3390/antibiotics12081295).
- [83] Wajid OM, Al-Kadmy I, Pourmand MR, Naderi G, Asadian M, Ghourchian S, Douraghi M. The context of blaOXA-23 gene in Iraqi carbapenem-resistant *Acinetobacter baumannii* isolates belonging to global clone 1 and global clone 2. *BMC Res Notes* 2024;17:300. <https://doi.org/10.1186/s13104-024-06890-w>.
- [84] Jeamsripong S, Thaotumpitak V, Anuntawirun S, Roongrojmongkhon N, Atwill ER, Hinthong W. Molecular epidemiology of antimicrobial resistance and virulence profiles of *Escherichia coli*, *Salmonella* spp., and vibrio spp. isolated from coastal seawater for aquaculture. *Antibiotics (Basel)* 2022;11:1688. <https://doi.org/10.3390/antibiotics11121688>.
- [85] Husna A, Rahman Md M, Badruzzaman ATM, Sikder MH, Islam MR, Rahman Md T, Alam J, Ashour HM. Extended-spectrum β -lactamases (ESBL): challenges and opportunities. *Biomedicine* 2023;11:2937. <https://doi.org/10.3390/biomedicine11112937>.
- [86] Teng J, Imani S, Zhou A, Zhao Y, Du L, Deng S, Li J, Wang Q. Combatting resistance: understanding multi-drug resistant pathogens in intensive care units. *Biomed Pharmacother* 2023;167:115564. <https://doi.org/10.1016/j.biopha.2023.115564>.
- [87] Pattolath A, Adhikari P, Pai V. Clinical and molecular profile of carbapenem resistant *Klebsiella pneumoniae* infections in a tertiary care hospital -Mangalore. *Infect Drug Resist* 2023;16:4335–48. <https://doi.org/10.2147/IDR.S411056>.
- [88] Mendes G, Santos ML, Ramalho JF, Duarte A, Caneiras C. Virulence factors in carbapenem-resistant hypervirulent *Klebsiella pneumoniae*. *Front Microbiol* 2023;14:1325077. <https://doi.org/10.3389/fmicb.2023.1325077>.
- [89] Ali AB, Al-Johani I, Al-Shamrani JM, Musamed AH, Al-Otaibi BG, Almazmomi K, Yusnouraini YN. Antimicrobial resistance in methicillin-resistant staphylococcus aureus. *Saudi J Biol Sci* 2023;30:103604. <https://doi.org/10.1016/j.sjbs.2023.103604>.
- [90] Malachowa N, DeLeo FR. Mobile genetic elements of *Staphylococcus aureus*. *Cell Mol Life Sci CMLS* 2010;67:3057–71. <https://doi.org/10.1007/s00018-010-0389-4>.
- [91] Miller WR, Munita JM, Arias CA. Mechanisms of antibiotic resistance in enterococci. *Expert Rev Anti Infect Ther* 2014;12:1221–36. <https://doi.org/10.1586/14787210.2014.956092>.
- [92] Lister PD, Wolter DJ, Hanson ND. Antibacterial-resistant *Pseudomonas aeruginosa*: clinical impact and complex regulation of chromosomally encoded resistance mechanisms. *Clin Microbiol Rev* 2009;22:582–610. <https://doi.org/10.1128/CMR.00040-09>.
- [93] Aguilar-Rodea P, Zúñiga G, Cerritos R, Rodríguez-Espino BA, Gomez-Ramirez U, Nolasco-Romero CG, López-Marcellano B, Rodea GE, Mendoza-Elizalde S, Reyes-López A, Olivares CH, Vigueras GJC, Velázquez-Guadarrama N, Rosas-Pérez I. Nucleotide substitutions in the mexR, nalC and nalD regulator genes of the MexAB-OprM efflux pump are maintained in *Pseudomonas aeruginosa* genetic lineages. *PLoS One* 2022;17:e0266742. <https://doi.org/10.1371/journal.pone.0266742>.
- [94] Wachino J-I, Doi Y, Arakawa Y. Aminoglycoside resistance: updates with a focus on acquired 16S ribosomal RNA methyltransferases. *Infect Dis Clin North Am* 2020;34:887–902. <https://doi.org/10.1016/j.idc.2020.06.002>.
- [95] Samreen AI, Malak HA, Abulreesh HH. Environmental antimicrobial resistance and its drivers: a potential threat to public health. *J Glob Antimicrob Resist* 2021;27:101–11. <https://doi.org/10.1016/j.jgar.2021.08.001>.
- [96] Kubde D, Badge AK, Ugemuge S, Shahu S. Importance of hospital infection control. *Cureus* 2023;15:e50931. <https://doi.org/10.7759/cureus.50931>.
- [97] Manyi-Loh C, Mamphweli S, Meyer E, Okoh A. Antibiotic use in agriculture and its consequential resistance in environmental sources: potential public health implications. *Molecules* 2018;23:795. <https://doi.org/10.3390/molecules23040795>.
- [98] Perez F, Endimiani A, Ray AJ, Decker BK, Wallace CJ, Hujer KM, Ecker DJ, Adams MD, Toltzis P, Dul MJ, Windau A, Bajaksouzian S, Jacobs MR, Salata RA, Bonomo RA. Carbapenem-resistant *Acinetobacter baumannii* and *Klebsiella pneumoniae* across a hospital system: impact of post-acute care facilities on dissemination. *J Antimicrob Chemother* 2010;65:1807–18. <https://doi.org/10.1093/jac/dkq191>.
- [99] van Duin D, Paterson DL. Multidrug-resistant bacteria in the community: an update. *Infect Dis Clin North Am* 2020;34:709–22. <https://doi.org/10.1016/j.idc.2020.08.002>.
- [100] Hedman HD, Vasco KA, Zhang L. A review of antimicrobial resistance in poultry farming within low-resource settings. *Anim Open Access J MDPI* 2020;10:1264. <https://doi.org/10.3390/ani10081264>.
- [101] Haenni M, Dagot C, Chesneau O, Bibbal D, Labanowski J, Viallette M, Bouchard D, Martin-Laurent F, Calsat L, Nazaret S, Petit F, Pourcher A-M, Togola A, Bachelot M, Topp E, Hocquet D. Environmental contamination in a high-income country (France) by antibiotics, antibiotic-resistant bacteria, and antibiotic resistance genes: status and possible causes. *Environ Int* 2022;159:107047. <https://doi.org/10.1016/j.envint.2021.107047>.
- [102] Caneschi A, Bardhi A, Barbarossa A, Zaghini A. The use of antibiotics and antimicrobial resistance in veterinary medicine, a complex phenomenon: a narrative review. *Antibiotics (Basel)* 2023;12:487. <https://doi.org/10.3390/antibiotics12030487>.
- [103] Salam Md A, Al-Amin Md Y, Salam MT, Pawar JS, Akhter N, Rabaan AA, Alqumber MAA. Antimicrobial resistance: a growing serious threat for global public health. *Healthcare* 2023;11:1946. <https://doi.org/10.3390/healthcare11131946>.
- [104] Zhou C, Jin L, Wang Q, Wang X, Chen F, Gao Y, Zhao C, Chen H, Cao B, Wang H. Bloodstream infections caused by carbapenem-resistant enterobacterales: risk factors for mortality, antimicrobial therapy and treatment outcomes from a prospective multicenter study. *Infect Drug Resist* 2021;14:731–42. <https://doi.org/10.2147/IDR.S294282>.
- [105] Hotchkiss RS, Moldawer LL, Opal SM, Reinhart K, Turnbull IR, Vincent J-L. Sepsis and septic shock. *Nat Rev Dis Prim* 2016;2:16045. <https://doi.org/10.1038/nrdp.2016.45>.
- [106] Llor C, Bjerrum L. Antimicrobial resistance: risk associated with antibiotic overuse and initiatives to reduce the problem. *Ther Adv Drug Saf* 2014;5:229–41. <https://doi.org/10.1177/2042098614554919>.
- [107] Lai C, Ma Z, Luo Y, Gao Y, Wu Z, Zhang J, Xu W. Factors influencing mortality in intracranial infections caused by carbapenem-resistant *Klebsiella pneumoniae*. *Sci Rep* 2024;14:20670. <https://doi.org/10.1038/s41598-024-71660-4>.
- [108] So-Ngern A, Osathai N, Meesing A, Chumpangern W. Mortality rate and factors associated with mortality of carbapenem-resistant Enterobacteriaceae infection. *Drug Target Insights* 2023;17:120–5. <https://doi.org/10.33393/dti.2023.2622>.
- [109] Theodorakis N, Feretzakis G, Hitas C, Kreouzi M, Kalantzi S, Spyridaki A, Boufeas IZ, Sakagianni A, Paxinou E, Vergykios VS, Nikolaou M. Antibiotic resistance in the elderly: mechanisms, risk factors, and solutions. *Microorganisms* 2024;12:1978. <https://doi.org/10.3390/microorganisms12101978>.
- [110] Weinberg SE, Villedieu A, Bagdasarian N, Karah N, Teare L, Elamin WF. Control and management of multidrug resistant *Acinetobacter baumannii*: a review of the evidence and proposal of novel approaches. *Infect Prev Pract* 2020;2:100077. <https://doi.org/10.1016/j.infpip.2020.100077>.
- [111] Poudel AN, Zhu S, Cooper N, Little P, Tarrant C, Hickman M, Yao G. The economic burden of antibiotic resistance: a systematic review and meta-analysis. *PLoS One* 2023;18:e0285170. <https://doi.org/10.1371/journal.pone.0285170>.
- [112] Weigelt JA, Lipsky BA, Tabak YP, Derby KG, Kim M, Gupta V. Surgical site infections: causative pathogens and associated outcomes. *Am J Infect Control* 2010;38:112–20. <https://doi.org/10.1016/j.ajic.2009.06.010>.
- [113] Almutairy B. Extensively and multidrug-resistant bacterial strains: case studies of antibiotics resistance. *Front Microbiol* 2024;15:1381511. <https://doi.org/10.3389/fmicb.2024.1381511>.
- [114] Bassetti M, Kanj SS, Kiratisin P, Rodrigues C, Van Duin D, Villegas MV, Yu Y. Early appropriate diagnostics and treatment of MDR gram-negative infections. *JAC Antimicrob Resist* 2022;4:dlac089. <https://doi.org/10.1093/jacamr/dlac089>.
- [115] Thacharodi A, Vithlani A, Hassan S, Alqahtani A, Pugazhendhi A. Carbapenem-resistant *Acinetobacter baumannii* raises global alarm for new antibiotic regimens. *iScience* 2024;27:111367. <https://doi.org/10.1016/j.isci.2024.111367>.
- [116] Sartelli M, C Hardcastle T, Catena F, Chichom-Mefire A, Coccolini F, Dhingra S, Haque M, Hodonou A, Iskandar K, Labricciosa FM, Marmorale C, Sall I, Pagani L. Antibiotic use in low and middle-income countries and the challenges of antimicrobial resistance in surgery. *Antibiotics (Basel)* 2020;9:497. <https://doi.org/10.3390/antibiotics9080497>.
- [117] Huemer M, Mairpady SS, Brugger SD, Zinkernagel AS. Antibiotic resistance and persistence—Implications for human health and treatment perspectives. *EMBO Rep* 2020;21:e51034. <https://doi.org/10.15252/embr.202051034>.
- [118] Pokharel S, Raut S, Adhikari B. Tackling antimicrobial resistance in low-income and middle-income countries. *BMJ Glob Health* 2019;4:e002104. <https://doi.org/10.1136/bmjgh-2019-002104>.
- [119] Yamin D, Uskoković V, Wakil AM, Goni MD, Shamsuddin SH, Mustafa FH, Alfouzan WA, Alissa M, Alshengeti A, Almaghrabi RH, Fares MAA, Garout M, Al Kaabi NA, Alshehri AA, Ali HM, Rabaan AA, Aldubisi FA, Yean CY, Yusof NY. Current and future technologies for the detection of antibiotic-resistant bacteria. *Diagnostics* 2023;13:3246. <https://doi.org/10.3390/diagnostics13203246>. Basel, Switzerland.
- [120] Zhang H, Yu F, Lu X, Li Y, Peng D, Wang Z, Liu Y. Rapid detection of MCR-mediated colistin resistance in *Escherichia coli*. *Microbiol Spectr* 2022;10:e0092022. <https://doi.org/10.1128/spectrum.00920-22>.
- [121] Abushanab D, Nasr ZG, Al-Badriyeh D. Efficacy and safety of Colistin versus Tigecycline for multi-drug-resistant and extensively drug-resistant gram-negative pathogens-A meta-analysis. *Antibiotics (Basel)* 2022;11:1630. <https://doi.org/10.3390/antibiotics11111630>.
- [122] Zeng M, Xia J, Zong Z, Shi Y, Ni Y, Hu F, Chen Y, Zhuo C, Hu B, Lv X, Li J, Liu Z, Zhang J, Yang W, Yang F, Yang Q, Zhou H, Li X, Wang J, Li Y, Ren J, Chen B, Chen D, Wu A, Guan X, Qu J, Wu D, Huang X, Qiu H, Xu Y, Yu Y, Wang M. Guidelines for the diagnosis, treatment, prevention and control of infections caused by carbapenem-resistant gram-negative bacilli. *J Microbiol Immunol Infect* 2023;56:653–71. <https://doi.org/10.1016/j.jmii.2023.01.017>.
- [123] Ali T, Habib A, Nazir Z, Ali M, Haque MA. Healthcare challenges in LMICs: addressing antibiotic resistance threats, a call for comprehensive global solutions: an editorial. *Int J Surg* 2024;110:3085–7. <https://doi.org/10.1097/JS9.0000000000001165>.
- [124] Aslam B, Asghar R, Muzammil S, Shafique M, Siddique AB, Khurshid M, Ijaz M, Rasool MH, Chaudhry TH, Aamir A, Baloch Z. AMR and sustainable development goals: at a crossroads. *Glob Health* 2024;20:73. <https://doi.org/10.1186/s12992-024-01046-8>.

- [125] McDonnell A., Countryman A., Laurence T., Gulliver S., Drake T., Edwards S., Kenny C., Lamberti O., Morton A., Shafira A., Smith R., Guzman J. 2024. Forecasting the fallout from AMR: economic impacts of antimicrobial resistance in humans.
- [126] Infectious Diseases Society of America (IDSA), Spellberg B, Blaser M, Guidos RJ, Boucher HW, Bradley JS, Eisenstein BI, Gerding D, Lynfield R, Reller LB, Rex J, Schwartz D, Septimus E, Tenover FC, Gilbert DN. Combating antimicrobial resistance: policy recommendations to save lives. *Clin Infect Dis* 2011;52(Suppl 5):S397–428. <https://doi.org/10.1093/cid/cir153>. : An Official Publication of the Infectious Diseases Society of America.
- [127] Seung KJ, Keshavjee S, Rich ML. Multidrug-resistant tuberculosis and extensively drug-resistant tuberculosis. *Cold Spring Harb Perspect Med* 2015;5:a017863. <https://doi.org/10.1101/cshperspect.a017863>.
- [128] Ajulo S, Awosile B. Global antimicrobial resistance and use surveillance system (GLASS 2022): investigating the relationship between antimicrobial resistance and antimicrobial consumption data across the participating countries. *PLoS One* 2024;19:e0297921. <https://doi.org/10.1371/journal.pone.0297921>.
- [129] Sharma J, Sharma D, Singh A, Sunita K. Colistin resistance and management of drug resistant infections. *Can J Infect Dis Med Microbiol* 2022;2022:4315030. <https://doi.org/10.1155/2022/4315030>. = *Journal Canadien Des Maladies Infectieuses Et De La Microbiologie Medicale*.
- [130] Global Antimicrobial Resistance and Use Surveillance System (GLASS). Available at <https://www.who.int/initiatives/glass> (accessed January 6, 2025).
- [131] Holm M, MacWright WR, Poudyal N, Shaw AG, Joh HS, Gallagher PF, Kim J-H, Shaikh AT, Seo HJ, Kwon SY, Priyati K, Dolabella B, Taylor BEW, Yeats C, Aanensen DM, Stelling J, Marks F. Capturing data on antimicrobial resistance patterns and trends in use in regions of Asia (CAPTURA). *Clin Infect Dis* 2023;77: S500–6. <https://doi.org/10.1093/cid/ciad567>. : An Official Publication of the Infectious Diseases Society of America.
- [132] Banerjee R, Humphries R. Clinical and laboratory considerations for the rapid detection of carbapenem-resistant Enterobacteriaceae. *Virulence* 2017;8:427–39. <https://doi.org/10.1080/21505594.2016.1185577>.
- [133] Seale AC, Hutchison C, Fernandes S, Stoesser N, Kelly H, Lowe B, Turner P, Hanson K, Chandler CIR, Goodman C, Stabler RA, Scott JAG. Supporting surveillance capacity for antimicrobial resistance: laboratory capacity strengthening for drug resistant infections in low and middle income countries. *Wellcome Open Res* 2017;2:91. <https://doi.org/10.12688/wellcomeopenres.12523.1>.
- [134] Eddiiteed P., 2017. Enquiries and Feedback direct all correspondence to: Cabinet Secretary Ministry of Health P. O. Box 30016 GPO Nairobi 00100.
- [135] Shrestha P, He S, Legido-Quigley H. Antimicrobial resistance research collaborations in Asia: challenges and opportunities to equitable partnerships. *Antibiotics (Basel)* 2022;11:755. <https://doi.org/10.3390/antibiotics11060755>.
- [136] Sartelli M, Marini CP, McNelis J, Coccolini F, Rizzo C, Labricciosa FM, Petrone P. Preventing and controlling healthcare-associated infections: the first principle of every antimicrobial stewardship program in hospital settings. *Antibiotics (Basel)* 2024;13:896. <https://doi.org/10.3390/antibiotics13090896>.
- [137] Shelke YP, Bankar NJ, Bandre GR, Hawale DV, Dawande P. An overview of preventive strategies and the role of various organizations in combating antimicrobial resistance. *Cureus* 2023;15:e44666. <https://doi.org/10.7759/cureus.44666>.
- [138] Allcock S, Young EH, Holmes M, Gurdasani D, Dougan G, Sandhu MS, Solomon L, Török ME. Antimicrobial resistance in human populations: challenges and opportunities. *Glob Health Epidemiol Genom* 2017;2:e4. <https://doi.org/10.1017/ghg.2017.4>.
- [139] Zavaleta-Monestel E, Hasselmyr HS, García-Montero J, Arguedas-Chacón S, Rojas-Chinchilla C, Díaz-Madriz JP. The impact of vaccination as a strategy to combat bacterial antimicrobial resistance. *Cureus* 2024;16:e65840. <https://doi.org/10.7759/cureus.65840>.
- [140] Wang H, Jia C, Li H, Yin R, Chen J, Li Y, Yue M. Paving the way for precise diagnostics of antimicrobial resistant bacteria. *Front Mol Biosci* 2022;9:976705. <https://doi.org/10.3389/fmolb.2022.976705>.
- [141] Aslam B, Wang W, Arshad MI, Khurshid M, Muzammil S, Rasool MH, Nisar MA, Alvi RF, Aslam MA, Qamar MU, Salamat MKF, Baloch Z. Antibiotic resistance: a rundown of a global crisis. *Infect Drug Resist* 2018;11:1645–58. <https://doi.org/10.2147/IDR.S173867>. B4691.pdf.
- [142] Ahmed MM, Kayode HH, Okesanya OJ, Ukoaka BM, Eshun G, Mourid MR, Adigun OA, Ogaya JB, Mohamed ZO, Lucero-Priso DE. CRISPR-Cas systems in the fight against antimicrobial resistance: current status, potentials, and future directions. *Infect Drug Resist* 2024;17:5229–45. <https://doi.org/10.2147/IDR.S494327>.
- [143] Ahmed SK, Hussein S, Qurbani K, Ibrahim RH, Fareeq A, Mahmood KA, Mohamed MG. Antimicrobial resistance: impacts, challenges, and future prospects. *J Med Surg Public Health* 2024;2:100081. <https://doi.org/10.1016/j.glmed.2024.100081>.
- [144] Dangor Z, Benson N, Berkley JA, Bielicki J, Bijsma MW, Broad J, Buurman ET, Cross A, Duffy EM, Holt KE, Iroh Tam P-Y, Jit M, Karampatsas K, Katwere M, Kwatra G, Laxminarayan R, Le Doare K, Mboizi R, Micoli F, Moore CE, Nakabembe E, Naylor NR, O'Brien S, Olwagen C, Reddy D, Rodrigues C, Rosen DA, Sadarangani M, Srikanthiah P, Tennant SM, Hasso-Agopsowicz M, Madhi SA. Vaccine value profile for Klebsiella pneumoniae. *Vaccine* 2024;42: S125–41. <https://doi.org/10.1016/j.vaccine.2024.02.072>.
- [145] Xuan J, Feng W, Wang J, Wang R, Zhang B, Bo L, Chen Z-S, Yang H, Sun L. Antimicrobial peptides for combating drug-resistant bacterial infections. *Drug Resist Updates* 2023;68:100954. <https://doi.org/10.1016/j.drug.2023.100954>.
- [146] Tookey CL, Hinchliffe P, Bragginton EC, Colenso CK, Hirvonen VHA, Takebayashi Y, Spencer J. β -lactamases and β -lactamase inhibitors in the 21st century. *J Mol Biol* 2019;431:3472–500. <https://doi.org/10.1016/j.jmb.2019.04.002>.
- [147] Murugaiyan J, Kumar PA, Rao GS, Iskandar K, Hawser S, Hays JP, Mohsen Y, Adukkadukkam S, Awuah WA, Jose RAM, Sylvia N, Nansubuga EP, Tiloca B, Roncada P, Roson-Calero N, Moreno-Morales J, Amin R, Kumar BK, Kumar A, Toufik A-R, Zaw TN, Akinwotu OO, Satyaseela MP, van Dongen MBM. Progress in alternative strategies to combat antimicrobial resistance: focus on antibiotics. *Antibiotics (Basel)* 2022;11:200. <https://doi.org/10.3390/antibiotics11020200>.
- [148] Belay WY, Getachew M, Tegegne BA, Teffera ZH, Dagne A, Zeleke TK, Abebe RB, Gedif AA, Fenta A, Yirdaw G, Tilahun A, Aschale Y. Mechanism of antibacterial resistance, strategies and next-generation antimicrobials to contain antimicrobial resistance: a review. *Front Pharmacol* 2024;15:1444781. <https://doi.org/10.3389/fphar.2024.1444781>.
- [149] Baran A, Kwiatkowska A, Potocki L. Antibiotics and bacterial resistance-A short story of an endless arms race. *Int J Mol Sci* 2023;24:5777. <https://doi.org/10.3390/ijms24065777>.
- [150] MacNair CR, Tsai CN, Rutherford ST, Tan M-W. Returning to nature for the next generation of antimicrobial therapeutics. *Antibiotics (Basel)* 2023;12:1267. <https://doi.org/10.3390/antibiotics12081267>.
- [151] Lin DM, Koskella B, Lin HC. Phage therapy: an alternative to antibiotics in the age of multi-drug resistance. *World J Gastrointest Pharmacol Ther* 2017;8:162–73. <https://doi.org/10.4292/wjgpt.v8.i3.162>.
- [152] Muteeb G. Nanotechnology—A light of hope for combating antibiotic resistance. *Microorganisms* 2023;11:1489. <https://doi.org/10.3390/microorganisms11061489>.
- [153] Cattoir V, Felden B. Future antibacterial strategies: from basic concepts to clinical challenges. *J Infect Dis* 2019;220:350–60. <https://doi.org/10.1093/infdis/jiz134>.
- [154] Hitchcock NM, Devequi GND, Shiach J, Valeria SHK, Dantas VBJ, Alencar PRL, Coler BS, Botelho PSM, Badaró R. Current clinical landscape and global potential of bacteriophage therapy. *Viruses* 2023;15:1020. <https://doi.org/10.3390/v15041020>.
- [155] Wang Y, Song M, Chang W. Antimicrobial peptides and proteins against drug-resistant pathogens. *Cell Surf* 2024;12:100135. <https://doi.org/10.1016/j.tcs.2024.100135>.
- [156] Xu Y, Li Z. CRISPR-Cas systems: overview, innovations and applications in human disease research and gene therapy. *Comput Struct Biotechnol J* 2020;18:2401–15. <https://doi.org/10.1016/j.csbj.2020.08.031>.
- [157] Navalkele BD, Chopra T. Bezlotoxumab: an emerging monoclonal antibody therapy for prevention of recurrent Clostridium difficile infection. *Biol Targets Therapy* 2018;12:11–21. <https://doi.org/10.2147/BTT.S127099>.
- [158] Langdon A, Schwartz DJ, Bulow C, Sun X, Hink T, Reske KA, Jones C, Burnham C-AD, Dubberke ER, Dantas G, CDC Prevention Epicenter Program. Microbiota restoration reduces antibiotic-resistant bacteria gut colonization in patients with recurrent Clostridioides difficile infection from the open-label PUNCH CD study. *Genome Med* 2021;13:28. <https://doi.org/10.1186/s13073-021-00843-9>.
- [159] Ahsan A, Thomas N, Barnes TJ, Subramaniam S, Loh TC, Joyce P, Prestidge CA. Lipid nanocarriers-enabled delivery of antibiotics and antimicrobial adjuvants to overcome bacterial biofilms. *Pharmaceutics* 2024;16:396. <https://doi.org/10.3390/pharmaceutics16030396>.
- [160] González-Bello C. Antibiotic adjuvants – a strategy to unlock bacterial resistance to antibiotics. *Bioorg Med Chem Lett* 2017;27:4221–8. <https://doi.org/10.1016/j.bmcl.2017.08.027>.
- [161] AlSheikh HMA, Sultan I, Kumar V, Rather IA, Al-Sheikh H, Tasleem JA, Haq QMR. Plant-based phytochemicals as possible alternative to antibiotics in combating bacterial drug resistance. *Antibiotics (Basel)* 2020;9:480. <https://doi.org/10.3390/antibiotics9080480>.
- [162] Ababneh MA, Nasser SA, Rababa'h AM. A systematic review of antimicrobial stewardship program implementation in Middle Eastern countries. *Int J Infect Dis* 2021;105:746–52. <https://doi.org/10.1016/j.ijid.2021.03.035>.
- [163] Antimicrobial stewardship programmes in health-care facilities in low- and middle-income countries: a WHO practical toolkit. *JAC Antimicrob Resist* 2019;1: dlz072. <https://doi.org/10.1093/jacamr/dlz072>.
- [164] Tinker NJ, Foster RA, Webb BJ, Haydoura S, Buckel WR, Stenehjem EA. Interventions to optimize antimicrobial stewardship. *Antimicrob Stewardsh Healthc Epidemiol ASHE* 2021;1:e46. <https://doi.org/10.1017/ash.2021.210>.
- [165] Mohammed AM, Mohammed M, Olewi JK, Osman AF, Adam T, Betar BO, Gopinath SCB, Ihmedee FH. Enhancing antimicrobial resistance strategies: leveraging artificial intelligence for improved outcomes. *S Afr J Chem Eng* 2025; 51:272–86. <https://doi.org/10.1016/j.sajce.2024.12.005>.
- [166] Worldwide Antimicrobial Resistance National/International Network Group (WARNING) Collaborators. Ten golden rules for optimal antibiotic use in hospital settings: the WARNING call to action. *World J Emerg Surg WJES* 2023;18:50. <https://doi.org/10.1186/s13017-023-00518-3>.
- [167] Bassetti S, Tschudin-Sutter S, Egli A, Osthoff M. Optimizing antibiotic therapies to reduce the risk of bacterial resistance. *Eur J Intern Med* 2022;99:7–12. <https://doi.org/10.1016/j.ejim.2022.01.029>.
- [168] Godman B, Egwuenu A, Haque M, Malande OO, Schellack N, Kumar S, Saleem Z, Sneddon J, Hoxha I, Islam S, Mwita J, do Nascimento RCRM, Dias GIP, Niba LL, Amu AA, Acolatse J, Incoom R, Sefah IA, Oponga S, Kurdi A, Chikowe I, Khuluza F, Kibuule D, Ogunleye OO, Olalekan A, Markovic-Pekovic V, Meyer JC, Alfiadi A, Phuong TNT, Kalungia AC, Campbell S, Pisana A, Wale J, Seaton RA. Strategies to improve antimicrobial utilization with a special focus on developing countries. *Life (Basel)* 2021;11:528. <https://doi.org/10.3390/life11060528>.

- [169] Micallef C, Enoch DA, Kamranpour P, Santos R, Tyler N, Scott S. The role of hospital antimicrobial and infectious diseases pharmacists in the UK: a theoretically underpinned exploration. *JAC Antimicrob Resist*, 5; 2023dlac136. <https://doi.org/10.1093/jacamr/dlac136>.
- [170] Rezel-Potts E, Gulliford M. Electronic Health records and antimicrobial stewardship research: a narrative review. *Curr Epidemiol Rep* 2022;1–10. <https://doi.org/10.1007/s40471-021-00278-1>.
- [171] Farooqui HH, Selvaraj S, Mehta A, Heymann DL. Community level antibiotic utilization in India and its comparison vis-à-vis European countries: evidence from pharmaceutical sales data. *PLoS One* 2018;13:e0204805. <https://doi.org/10.1371/journal.pone.0204805>.
- [172] Koly KN, Baskin C, Khanam I, Rao M, Rasheed S, Law GR, Sarker F, Gnani S. Educational and training interventions aimed at healthcare workers in the detection and management of people with mental health conditions in South and South-East Asia: a systematic review. *Front Psychiatry* 2021;12:741328. <https://doi.org/10.3389/fpsy.2021.741328>.
- [173] Chalkidou A, Lambert M, Cordoba G, Taxis K, Hansen MP, Bjerrum L. Misconceptions and knowledge gaps on antibiotic use and resistance in four healthcare settings and five European countries-a modified Delphi study. *Antibiotics (Basel)* 2023;12:1435. <https://doi.org/10.3390/antibiotics12091435>.
- [174] Vitale Brovarone E, Cotella G. Improving rural accessibility: a multilayer approach. *Sustainability* 2020;12:2876. <https://doi.org/10.3390/su12072876>.
- [175] Fanelli U, Chiné V, Pappalardo M, Gismondi P, Esposito S. Improving the quality of hospital antibiotic use: impact on multidrug-resistant bacterial infections in children. *Front Pharmacol* 2020;11:745. <https://doi.org/10.3389/fphar.2020.00745>.
- [176] Kamyab H, Khademi T, Chelliapan S, SaberiKamarposhti M, Rezaia S, Yusuf M, Farajnezhad M, Abbas M, Hun JB, Ahn Y. The latest innovative avenues for the utilization of artificial intelligence and big data analytics in water resource management. *Results Eng* 2023;20:101566. <https://doi.org/10.1016/j.rineng.2023.101566>.
- [177] Gilham EL, Pearce-Smith N, Carter V, Ashiru-Oredope D. Assessment of global antimicrobial resistance campaigns conducted to improve public awareness and antimicrobial use behaviours: a rapid systematic review. *BMC Public Health* 2024;24:396. <https://doi.org/10.1186/s12889-024-17766-w>.
- [178] Heidt B, Siqueira WF, Eersels K, Diliën H, van Grinsven B, Fujiwara RT, Cleij TJ. Point of care diagnostics in resource-limited settings: a review of the present and future of PoC in its most needed environment. *Biosensors (Basel)* 2020;10:133. <https://doi.org/10.3390/bios10100133>.
- [179] Alowais SA, Alghamdi SS, Alsuhebany N, Alqahtani T, Alshaya AI, Almohareb SN, Aldairem A, Alrashed M, Bin Saleh K, Badreldin HA, Al Yami MS, Al Harbi S, Albekairy AM. Revolutionizing healthcare: the role of artificial intelligence in clinical practice. *BMC Med Educ* 2023;23:689. <https://doi.org/10.1186/s12909-023-04698-z>.